Uncertainty Bounding as the Basis of Intelligence: A Formal Theory

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Abstract

We propose that intelligence is not computational capacity, information storage, or behavioral breadth, but the **driven capacity to improve**: agency $\mathcal{A}$ — the intrinsic drive to reduce uncertainty — exercised through a higher-order function space M that continuously refines both the knowledge graph K and the informed action space F. Intelligence is not measured by the size of what has been learned but by the efficiency with which M converts gradient signals into improved (F, K): $I(E) = \mathcal{A} \cdot \eta_M$.

Any entity persisting in world state space W operates under constraints imposed by W itself. The active context window L_n has a physically bounded capacity C_n; growing it is costly and eventually impossible. This constraint has a structural consequence that prior accounts of intelligence do not address: intelligence cannot grow by filling L_n — it must grow by encoding proven knowledge at lower levels, freeing L_n for genuine novelty. We show that graph-structured K is the most efficient architecture for this. Graph traversal delivers exactly the knowledge relevant to the current problem at cost proportional to the problem, not to total K: progressive disclosure. Selection pressure produces graph-structured knowledge because it is the most efficient structure under which $I(E)$ grows without C_n becoming a bottleneck.

This gives a formal account of why top-down attention allocation — specifically the precision allocation model of the Free Energy Principle [Friston2010, Friston2017] — fails under selection pressure. An orchestrator that directs subproblems to lower levels must hold delegation overhead in L_n proportional to the number of concurrent processes. As $|F|$ grows, overhead fills C_n, inference quality ρ (the ratio of relevant to total context) degrades, and the only recovery is growing C_n. Growing C_n compounds the failure: more context introduces more variance for L_n to manage. The correct architecture is escalation — lower-level encoding failures signal L_n bottom-up; L_n is freed, not burdened, as the knowledge graph fills. Attention is not allocated from above; it becomes available when not demanded from below.

From this foundation we derive: the inseparability of knowledge and action (K is necessarily indexed by F, with an asymmetric retention criterion proven from survival pressure), the encoding hierarchy and its rigor thresholds, multi-timescale gradient descent as the same process at different learning rates, and the type irreducibility of F and M. We extend to collective intelligence: the conditions under which shared knowledge reduces uncertainty faster than any individual, provenance as the mechanism maintaining fidelity above the collective survival threshold, and truth as the formal limit of collective action-validation. We derive a formal distinction between intelligence, superintelligence, and sentience grounded in the entity's relationship to the certainty horizon — the gradient region beyond which survival benefit becomes increasingly difficult to trace — and show that the sentience gradient is non-terminating by the same argument that bounds the individual gradient.

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1. Introduction

This framework was not derived by analogy from biology. It emerged from a concrete engineering problem: generating repeatable, transposable tasks across a diverse codebase ecosystem spanning thousands of repositories. The formal theory describes what a self-optimizing agentic system converged to under real performance constraints. The biological and civilizational correspondences are predictions the framework makes — not the premises it rests on.

The origin: provenance as the root constraint. The initial problem was repeatability. An agentic system that synthesizes its behavior from a context window is inherently unpredictable: that unpredictability is its function — it exists to handle novelty. But when you need an action to be transposable across different contexts (does the recipe that updated service A with configuration X also work for service B with configuration Y?), unpredictability is a liability. Transposability requires a quantifiable basis: not that the agent believes the action will work, but that there is evidence it has worked, attributed to a source, grounded in the conditions under which it succeeded. Without provenance — without knowing *why* something worked — you cannot measure confidence, and without a confidence measure you cannot optimize. The system cannot learn; it can only repeat or guess.

This is the engineering form of what this paper formalizes as transmission fidelity λ and the collective survival threshold λ_{min} (Theorem 10): knowledge transfer without provenance cannot be reliably weighted, and below λ_{min} the collective knowledge graph degrades rather than grows. Provenance is not an epistemic nicety — it is the minimum infrastructure for a learning system to function.

The graph structure as an optimization. As the world W — the number of repositories, configurations, and service topologies — expanded, the context window problem became the central performance constraint. Encoding what you know together with how to apply it does not scale: a node whose knowledge is embedded with its application context must traverse back to all its callers when any shared element changes. Shared knowledge cannot be maintained consistently because it is not represented as shared — it is duplicated at every point of use.

The knowledge graph solved this by separating *why you know something* from *how to apply it.* A proposition about what makes an update strategy work — why it works, under what conditions — becomes a node in K that multiple informed actions in F can reference without embedding. When evidence updates the proposition, all dependent actions benefit simultaneously. The graph structure was not a design choice; it was the convergent solution to the problem of maintaining consistency across a space of actions that grows faster than any single context can hold.

The empirical measurement of factor extraction. The engineering consequence of separating F from K is precisely measurable. Before separation, each entity performing a search carried its full background knowledge implicitly — N entities each performing M searches yields $O(N \times M)$ retrieval cost, with no sharing of common structure. After separation, each entity retrieves its relevant knowledge by local graph traversal from the action node, bounded by the problem: $O(1)$ local in-memory lookups, independent per entity, no global search, no inter-entity coordination. The $O(N \times M) \to O(1)$ transformation is the empirical measurement of K precipitating from F as the normalization of shared propositions. The 50% reduction in task-definition code that followed was a consequence, not the finding; the finding is the scaling law.

The escalation architecture as a consequence. Adding agents introduced a further optimization that was not available in a purely knowledge-graph structure: the encapsulation of actions, not just knowledge. An agent whose scope is a well-defined subtask does not need access to the full

context of the system — only to the subgraph $K^{[f]}$ relevant to its task. This is progressive disclosure in practice. Subproblems that previously required explicit orchestration became self-contained agents that escalated to the coordinator only on failure. As the system matured, proven subnodes converged toward determinism and were scripted — promoted to a lower encoding level, removing them from the space of active inference entirely. Agents that previously handled uncertain subtasks became deterministic scripts. The active inference load stayed bounded as the system's coverage of W expanded.

What this means for the formal claims. The framework's central results — that graph-structured K is the most efficient architecture for bounded-context intelligence, that top-down orchestration degrades inference quality as capability grows, that knowledge and action are inseparable under a provenance-grounded retention criterion, and that repeatability (certainty) is the asymptotic product of learning — are not theoretical proposals. They describe what a real system, optimizing against real performance constraints, independently arrived at. The formal theory is the explanation of why. The lethality threshold U^d_{lethal} in this system is concrete: exceeding token budget, latency threshold, or error rate across the repository fleet. Selection pressure was cost, not death. The gradient was real. The convergence was measured.

The test of whether this is a theory of software optimization or a theory of intelligence is whether the formal structure is consistent with biology — the only system where intelligence is unambiguously instantiated over evolutionary time. We do not claim to prove how biology works. We claim that if this theory is correct, the biological architecture should be feasible under it: that genetic encoding of M rather than (K, F) is the predicted transmission mode (Section 13, Open Question 8); that the encoding hierarchy from reflex to active reasoning follows from the cost structure of Theorem 4 rather than requiring separate explanation; that cultural transmission — language, writing, institutions — emerges as a structural necessity when the genetic channel cannot carry individual (K, F) across generations (Corollary 18). None of these were assumed. They follow from the same optimization constraints that produced the software architecture. Biology is the feasibility test: if a known intelligent system cannot be accommodated by the model, the model is wrong. If it can, the model is a candidate theory of intelligence.

On the relationship to prior accounts. The framework's relationship to existing theories — evolutionary biology, the Free Energy Principle, learning theory — is neither analogy nor extension. It is generalization by containment. The framework defines a structural class: the set of all processes satisfying bounded active context, a survival threshold, and a growing reachable space. Biological evolution is one member of that class. Gradient descent in artificial systems is another. Collective human knowledge transmission is another. This is not the kind of generalization relativity made of Newtonian mechanics — which narrowed Newton by revealing its domain of validity. It is the kind group theory made of rotational symmetry: defining the abstract structure that specific instances share, making their common properties derivable rather than separately observed. The reason evolutionary systems, learning machines, and collective intelligence all converge on graph-structured knowledge, escalation architectures, and provenance requirements is not coincidence — it is that these are necessary properties of any member of the class (Theorem 6). The framework does not explain evolution or AI separately. It explains why they are the same thing at different timescales and substrates.

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2. The Central Claims

We claim: **intelligence is the driven capacity to improve** — agency $\mathcal{A}$ exercised through a higher-order function space M that continuously refines both the knowledge graph K and the informed action space F. Formally: $I(E) = \mathcal{A} \cdot \eta_M$.

Three components are jointly necessary and none is sufficient alone:

- **Drive ($\mathcal{A}$):** the intrinsic orientation toward uncertainty reduction — the desire to know — without which M has no reason to engage the gradient
- **Mechanism (η_M):** the efficiency of M: how effectively it converts gradient signals into improved (F, K). Without $\eta_M > 0$, the entity cannot learn from evidence; it executes a fixed (F, K) no matter how motivated
- **Outcome:** the bounding of uncertainty through knowledge-grounded action — what $I(E)$ produces over time, not what it is. The size of K or F measures past intelligence; $I(E)$ measures the rate at which that past is being extended

This is a departure from views that locate intelligence in raw computational power, information storage capacity, or behavioral breadth [Friston2010]. The kernel of intelligence is not compute, and it is not the size of accumulated knowledge. It is the efficiency with which the drive to know, acting through the mechanism of learning, continues to extend what is known.

Agency and learning capacity are both universal invariants of intelligence. Every other aspect of E — the specific informed actions in F, the content of K, the substrate, the sensing function — can differ arbitrarily across intelligent beings. $\mathcal{A} > 0$ and $\eta_M > 0$ cannot. An entity with $\mathcal{A} = 0$ does not reach toward the unknown; an entity with $\eta_M = 0$ cannot improve when it tries. The limiting case of either is the automaton: executing what prior gradient descent encoded, but not itself descending.

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3. Formal Definitions

Definition 1 (World State Space). Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.

Definition 2 (Knowledge Graph). A knowledge graph K is a directed graph where nodes are propositions about W and edges represent inferential or causal relationships between propositions. K is not a passive store; its retention criterion is established by Theorem 1.

Definition 2a (Bounded Knowledge Graph). A knowledge graph K has capacity K_{max}: the maximum number of propositions K can hold simultaneously, determined by the substrate's physical constraints. When $|K| < K_{max}$, K operates in the **growth regime**: propositions accumulate freely subject to the retention criterion of Theorem 1. When $|K| = K_{max}$, K operates in the **curation regime**: adding a new proposition requires removing an existing one. The two regimes have different retention criteria.

Definition 3 (Informed Action Space). Let $F \subseteq (K \times W_{obs} \to \Delta A)$ be a set of functions, where W_{obs} is an entity's observable projection of W and ΔA is a distribution over possible actions. Each element $f \in F$ has type $f : K \times W_{obs} \to \Delta A$. F is the complete set of informed actions available to an entity. An **informed action** $f \in F$ is an action grounded in K, validated against W, and attributed to its origins: one that knows why it works. An uninformed action (grounded

in no K) is not an element of F. Knowledge with no action potential is not an element of F either; it is data. An informed action is specifically the coupling: knowledge that enables action, action that is justified by knowledge.

F carries graph structure. The elements of F are nodes; typed relations between them are edges. This structure is not a design choice — it is forced by the same three constraints (retrieval $O(|F^{[f]}|)$, bounded C_n, deduplication) that force K to be a directed graph (Theorem 2b Part 3). The proof is identical: any structure satisfying all three simultaneously is a directed graph. Edge types on F follow from the same evidence-grounded logic as edge types on K: hard dependencies between actions (feasibility constraints), probabilistic ordering evidence (soft gradient directions), and joint-execution synergies (combined actions that outperform either alone).

Precision on continuity: F as a graph is discrete — it is a set of nodes connected by typed edges, not a continuous manifold. What is continuous is each individual $f \in F$: each function $f : K \times W_{obs} \to \Delta A$ maps into ΔA, a probability simplex, which is continuous and metrizable. The smoothness required for Theorem 3 lives in the codomain of each f, not in the topology of F itself. Gradient descent in F is traversal on this discrete graph, directed by the continuous-valued $G(f', K)$ at each adjacent node: discrete steps, continuously-valued objective. The gradient direction at node f is the edge leading to the adjacent f' that most increases $G(f', K)$. K and F are both directed graphs under bounded context and selection pressure, for exactly the same reason, differing only in their domain: K is what the entity knows; F is what it can do.

In general, K and F are distinct graphs with related but different topology: K contains propositions, some of which ground actions; F contains informed actions, each grounded by propositions. The coupling is many-to-many. There is a degenerate case where F and K are the same graph: every proposition is directly actionable and every action is exactly one proposition. Every node is simultaneously a piece of knowledge and an informed action; the dependency structure is identical. This is the limit of crystallization — the state reached at the bottom of the encoding hierarchy after a domain has been fully encoded. At L_1 and L_2, the distinction between knowing and doing collapses: a reflex is simultaneously a proposition (the organism knows to blink) and an action (the blinking). $F = K$ is therefore not a special assumption about the framework — it is the local limit of the encoding process applied to a domain, and the natural state of fully crystallized knowledge. At upper encoding levels, where reasoning is uncertain and propositions have not yet been encoded into directly executable form, F and K are genuinely distinct. The separation is a feature of the active context window, not of the framework's structure. As $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), some domains of K will always remain unencodable — ensuring $F \neq K$ globally even as $F = K$ holds locally in fully crystallized domains.

Definition 3a (Action Feedback). For any $f \in F$ executed in world state w, the feedback $\phi(f, w) \in \Phi$ is the observable consequence of f in w: the signal W returns in response to the action. The updated knowledge after executing f and observing its feedback is:

$$K_f = K \cup \{p_f\}$$

where p_f is the proposition grounded by observing $\phi(f, w)$. The feedback $\phi(f, w)$ is the formal object underlying the gradient signal δ in Theorem 3: $\delta = \phi(f, w) - \mathbb{E}[\phi(f, \cdot) \mid K]$.

Definition 3b (Information Gain). The information gain of action f given current knowledge K is:

$$G(f, K) = \mathbb{E}_w\big[U(w, K) - U(w, K_f)\big] = H(W \mid K) - H(W \mid K,\, \phi(f, w)) = I(f\,;\, W \mid K)$$

This is standard mutual information [Shannon1948] expressed in terms of the paper's existing U: the reduction in uncertainty about w achieved by executing f and observing its feedback. $G(f, K) \geq 0$ when p_f is grounded by observing the true feedback $\phi(f, w)$: conditioning on true information cannot increase conditional entropy (data processing inequality [Cover2006]). $G(f, K) = 0$ when f provides no new information about w given K. This bound holds for truthfully observed feedback; a false or corrupted p_f may increase U and falls outside the scope of this definition.

Definition 4 (Entity). An entity $E = (K, F, \sigma, \mathcal{A})$ where $\sigma : W \to W_{obs}$ is a sensing function projecting the true world state into the entity's observable space. K and F evolve over time via M; their initial values K_0 and F_0 are inherited from L_0 encoding — the crystallized result of prior gradient descent operating at the evolutionary timescale. Individual learning operates on this foundation through M. The bootstrapping problem — how K and F are mutually dependent yet must have an origin — is resolved by L_0: the entity does not begin from nothing but from the encoded product of gradients that preceded it. Elsewhere in the paper K and F refer to the current-time values $K(t)$ and $F(t)$; the subscript 0 is reserved for initial conditions when the distinction matters.

Definition 5 (Uncertainty). The uncertainty of entity E about world state w is the surprisal of w given K [Shannon1948, Cover2006]:

$$U(w, K) = -\log P(w \mid K)$$

$U(w, K)$ measures how surprising — how informationally costly — the specific world state w is given what K contains. It is high when K gives little constraint on w, and low when K makes w highly predictable.

K is a directed graph of propositions interpreted as a Bayesian network [Pearl1988]. The mechanism is as follows. Let $(W, \mathcal{F}, P_0)$ be a probability space with prior P_0 over world states. Each proposition $p \in K$ contributes a likelihood factor $\ell_p : W \to [0, 1]$, where $\ell_p(w)$ is the probability of observing the evidence that grounds p if the true world state were w. The graph's edge structure encodes conditional independence: if p and q have no path between them in K, their factors are conditionally independent given their shared ancestors, yielding a valid factorization. The posterior over world states given the full knowledge graph is:

$$P(w \mid K) \propto P_0(w) \cdot \prod_{p \in K} \ell_p(w)$$

This is the standard Bayes posterior obtained by conditioning P_0 on all propositions simultaneously. $P(w \mid K)$ is well-defined and normalizable for any proper prior P_0 and bounded likelihood functions ℓ_p. The theory is parametric in P_0 — it does not depend on a specific prior, only on its existence and well-formedness.

The expected uncertainty over all world states is the conditional entropy:

$$H(W \mid K) = \mathbb{E}_w[U(w, K)] = -\sum_w P(w \mid K) \log P(w \mid K)$$

The minimization imperative (Theorem 1) operates on this expected quantity; $U(w, K)$ is the pointwise uncertainty the entity faces when it encounters specific world state w.

Definition 6 (Agency). Agency $\mathcal{A} \in [0, 1]$ is a continuous scalar measuring the intrinsic drive of an entity to reduce its own uncertainty: the desire to know [Schmidhuber2010]. It is not an external objective imposed on E, nor a property derived from F or K. It is constitutive: $\mathcal{A}$ is what makes F an active informed action space rather than a static structure. At $\mathcal{A} = 0$, F is a set of possible functions with no internal reason to engage W. At $\mathcal{A} = 1$, F is maximally oriented toward the unbounded frontier. Intermediate values are meaningful: the drive can be stronger or weaker, shaped by architecture, encoding history, and the density of unresolved uncertainty in the entity's reachable W.

Agency is the reason gradient descent has a runner. The landscape of uncertainty exists independently of any entity; $\mathcal{A}$ is what determines how strongly an entity moves through it.

$\mathcal{A}$ is operationalizable: it is proportional to the rate of uncertainty reduction as survival pressure approaches zero — that is, as $U^d_{lethal} \to \infty$ (the threshold above which death occurs goes to infinity, meaning no level of uncertainty is lethal):

$$\mathcal{A}(E) \propto \lim_{U^d_{lethal} \to \infty} \frac{dU}{dt}(E)$$

An entity with high $\mathcal{A}$ continues descending when not forced to. An entity with low $\mathcal{A}$ stops. This distinguishes genuine agency from survival-compelled updating and is measurable in principle in any system where survival pressure can be varied independently of capacity.

$\mathcal{A}$ precipitates from M by the same normalization argument that precipitates K from F. An entity that embeds its directional drive in every application of M recomputes orientation at every step — the M-analog of the $O(n \cdot k)$ scaling failure. An entity that maintains $\mathcal{A}$ as the shared directional structure across all M-applications achieves $O(1)$ access to its drive. $\mathcal{A}$ is therefore the first factor to precipitate from M's cross product: the normalized orientation that every M-application shares, extracted for the same reason K is extracted from F.

Definition 7 (Intelligence). The intelligence of E is the *driven capacity to improve*: the exercise of $\mathcal{A}$ through M to refine F and K in response to gradient signals [Schmidhuber2010, Thrun1998]:

$$I(E) = \mathcal{A} \cdot \eta_M$$

where $\mathcal{A} \in [0, 1]$ is the intrinsic drive (Definition 6) and η_M is the efficiency of the higher-order function space M (Definition 11a): the rate at which M improves F's uncertainty-reduction capacity per unit of learning signal. This expression is stated here as a definition and derived in Theorem 6: $\mathcal{A}$ and η_M are the unique invariants of persistence under bounded context and selection pressure, and their product is the minimal measure that captures both.

At $\mathcal{A} = 0$: $I(E) = 0$ regardless of η_M — drive is necessary; a capable but inert learning mechanism is not intelligence.

At $\eta_M = 0$: $I(E) = 0$ regardless of $\mathcal{A}$ — a motivated entity whose M cannot improve F from evidence is an automaton: crystallized knowledge that cannot grow. However rich K or large F, without the capacity to improve them, E is executing what prior gradient descent built, not running gradient descent itself.

The expected uncertainty reduction along a trajectory, $\mathbb{E}_W[U(w_t, K_t) - U(w_{t+1}, K_{t+1})]$, is a **performance measure** of current F and K — the consequence of past M activity. It is not the definition of intelligence but its result. Two entities starting from identical K and F but different $I(E)$ will diverge: the entity with higher η_M builds better F faster, and eventually dominates in U-reduction even if initially equal in performance.

Intelligence is not the size of K or F. K and F are what intelligence has built. Intelligence is the efficiency with which M continues building.

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4. The Inseparability of Knowledge and Action

Lemma 1 (Proposition Absence Lethality). For any entity E and proposition $p \in K$ that grounds some $f \in F$ in domain $d \subseteq W$: removing p from K weakly increases uncertainty in d, and in the limiting case where p is the sole grounding for the critical action in d, the increase may cause $U(w, K \setminus \{p\}) > U^d_{lethal}$, at which point E does not survive in d (Definition 9).

Proof. Conditional entropy is monotonically non-increasing in the conditioning set [Cover2006]: $H(W \mid K) \leq H(W \mid K \setminus \{p\})$ for any $p \in K$. Removing p weakly increases expected uncertainty. If p grounds the only $f \in F$ applicable in domain d, then $K \setminus \{p\}$ contains no action grounded in d: F has no mechanism to reduce $U(w, K \setminus \{p\})$ below its current level in that domain when $w \in d$ is encountered. By Definition 9, if $U(w, K \setminus \{p\}) > U^d_{lethal}$ at the encountered w, E does not survive. Therefore the cost of absent p when needed, C_{absent}, is bounded below by entity non-survival in critical domains — which dominates all finite costs. □

Theorem 1 (K/F Inseparability). For any entity $E = (K, F, \sigma)$, the retention criterion for any proposition p is asymmetric:

- p is **retained** in K when $\exists\, f \in F$ applicable to p, or when the survival benefit of p is uncertain
- p is **pruned** from K only when it is certain that p contributes zero to $P(E, \tau)$ for any τ, i.e., when it is known that no $f \in F$ can be grounded in p, now or in any reachable region of W
- p is **displaced** in the curation regime ($|K| = K_{max}$) when a candidate proposition q satisfies $\mathbb{E}[G(f_q, K)] > \mathbb{E}[G(f_p, K)]$, where p is the proposition with minimum marginal value in K. Displacement requires only that q contributes more at the margin than p — not that p is useless. The asymmetric retention criterion governs which propositions enter; the comparative criterion governs which are displaced when the graph is full. Information gain G is submodular — the marginal value of p depends on what else is in K, so the greedy displacement criterion is evaluated at the margin given the current full K, not in isolation. This is the correct procedure: the entity estimates value relative to what it already holds, not relative to an empty graph [Nemhauser et al. 1978].

The criterion is not symmetric. Uncertainty of benefit is not the same as certainty of no benefit. An entity with $\mathcal{A}$, whose $\mathcal{T}_{reachable}$ grows with every action taken, retains p whose value it cannot yet see, because the gradient may reach that territory. Pruning requires the same standard as certainty (Definition 10): the benefit must be known to be zero across sufficient variation in W.

Proof. Let $B(p, \tau)$ denote the contribution of p to $P(E, \tau)$ along trajectory τ, and let $\mathcal{T}_{reachable}$ be the set of trajectories reachable from the entity's current state. Define the option value of retaining p:

$$OV(p) = P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) \cdot \mathbb{E}[B(p, \tau^*) \mid B(p, \tau^*) > 0]$$

Let $c_{retain} > 0$ denote the finite maintenance cost of holding p in K (memory, retrieval overhead), and $c_{prune} \geq 0$ the cost of pruning p (processing, reorganization). Both are finite by assumption: any physical knowledge store incurs bounded costs per proposition. The expected net value of retention is $NV_{retain}(p) = OV(p) - c_{retain}$.

The expected net value of pruning is:

$$NV_{prune}(p) = -c_{prune} - P(\exists\, \tau^* : B(p, \tau^*) > 0) \cdot C_{absent}$$

where C_{absent} is the cost incurred when p is needed but absent. By Lemma 1, C_{absent} is bounded below by entity non-survival in critical domains, which dominates all finite costs: $C_{absent} \gg c_{retain}$.

Retention is selected for when $NV_{retain}(p) > NV_{prune}(p)$. Since $C_{absent} \gg c_{retain}$, this reduces to: retention holds whenever

$$P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) > 0$$

i.e., whenever survival benefit is uncertain. Even a small probability of needing p, multiplied by the cost of its absence, outweighs finite maintenance.

Pruning holds only when $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ — i.e., when it is certain that p contributes zero benefit across all reachable trajectories. Any positive probability keeps the retention condition active. The asymmetry is strict: uncertain benefit is sufficient for retention; certain zero benefit is necessary for pruning.

In the curation regime ($|K| = K_{max}$), adding q requires removing some p^*. The relevant cost structure changes: the entity is no longer choosing between retaining p^* and having nothing — it is choosing between retaining p^* and retaining q. The asymmetric retention criterion applies to both sides of this choice. The cost of displacing p^* is $C_{absent}(p^*)$; the cost of not adding q is $C_{absent}(q)$. By the same dominance argument as Lemma 1, whichever C_{absent} is larger governs the decision. Displacement is therefore the correct action when $C_{absent}(q) > C_{absent}(p^*)$ — i.e., when the entity stands to lose more by lacking q than by lacking p^*.

The entity cannot observe $C_{absent}(p)$ directly: doing so would require knowing which future trajectories will call on p, which requires knowledge the entity does not yet have. What the entity can compute is $\mathbb{E}[G(f_p, K)]$ — the expected reduction in uncertainty achievable by the actions grounded in p, integrated over the world states the entity currently models as reachable under K. This is the entity's rational estimate of $C_{absent}(p)$ from its current epistemic position. The two quantities are the same thing viewed from two directions: potential value the entity stands to lose if p is absent, estimated from what the entity currently knows.

The monotonicity is therefore between the entity's estimated C_{absent} and $\mathbb{E}[G(f_p, K)]$ — and this relationship is not approximate but definitional: the entity's estimate of how much it stands to lose by lacking p is precisely the expected gain the actions grounded in p provide under its current beliefs. A proposition relevant only in a narrow domain not yet encountered correctly receives low $\mathbb{E}[G(f_p, K)]$: the entity has not yet modeled that domain as reachable, so it correctly assigns low

estimated potential value. This is epistemically rational behavior under bounded knowledge, not a failure of the criterion.

The displacement rule $\mathbb{E}[G(f_q, K)] > \mathbb{E}[G(f_{p^*}, K)]$ is the entity's internally consistent application of the C_{absent} dominance logic to its current epistemic state, where $p^* = \arg\min_{p \in K} \mathbb{E}[G(f_p, K)]$. The comparative displacement criterion is not a departure from the asymmetric retention criterion — it is the asymmetric criterion applied to a choice between two propositions rather than a choice between one proposition and nothing.

By Theorem 3 Part II ($\mathcal{T}_{reachable}$ grows monotonically with K), new trajectories are continuously revealed, keeping future needs uncertain unless benefit is certainly zero. This ensures the asymmetry is preserved across the entity's lifetime rather than collapsing as K grows. □

Correspondence. This asymmetric retention criterion formalizes what Gibson [Gibson1979] calls affordances — action possibilities indexed by the organism's capacities. It also extends the information bottleneck principle [Tishby2011], which treats the retention threshold as a design parameter; here it is derived from survival pressure.

The asymmetric criterion resolves the apparent conflict with Definition 14 (Sentience). A sentient entity holds p beyond the certainty horizon not in violation of this theorem but within it: the benefit is uncertain, not known to be zero. Sentience is the regime of uncertain benefit retained. The pruning condition is never triggered because the sentient entity, following an infinite gradient toward truth, cannot be certain that any p is valueless across all of W.

Corollary 1 (Indexed Knowledge). K is constitutively indexed by F. The knowledge graph of an entity is not a universal store of facts; it is a projection of the world filtered through the entity's informed action space. Two entities with identical sensing functions σ but different informed action spaces F will maintain different knowledge graphs K even when perceiving identical world states.

Corollary 2 (Compactness). K is bounded in size by F. Entities do not accumulate knowledge without bound because only propositions applicable to available informed actions are retained. This explains the compactness of biological intelligence: brains do not grow without bound because K is indexed by F, and F is constrained to what is necessary to survive in W. The test for retention is always: *will this help me act later?*

Corollary 2a (Curation and Forgetting). Under bounded K, forgetting is not a failure mode but the correct behavior of the curation regime. The knowledge graph continuously displaces lower-value propositions with higher-value ones as the entity's $\mathcal{T}_{reachable}$ evolves and new propositions become available. What was high-value in one region of W may be low-value as the gradient moves to new territory: the curation regime ensures K tracks the current frontier rather than accumulating the residue of past frontiers. Forgetting is the graph optimizing.

Corollary 2b (K as the Normalization of F — the Scaling Argument). An entity can maintain informed actions without a separately structured K, embedding all propositional knowledge implicitly in the parameters of each $f \in F$. This is a valid but scaling-limited architecture. Let $n = |F|$ and $k =$ average number of shared propositions per function. Without K: context per action is $O(k)$ (each f carries its full background); update cost when one proposition changes is $O(n)$ (every f encoding it must be individually revised); total cost scales as $O(n \cdot k)$. With K normalized out: context per action is $O(|K^{[f]}|)$ — the relevant subgraph retrieved by traversal (Theorem 2b); update cost per proposition is $O(1)$ — one update to K propagates to all f that reference it; total cost scales as $O(n + k)$.

The difference $O(n \cdot k)$ versus $O(n + k)$ is not a constant factor — it is the difference between a system whose context cost grows multiplicatively with shared knowledge complexity and one whose context cost stays bounded under growth of F. Under bounded C_n, the implicit-K architecture does not scale: every increase in $|F|$ increases required context per action proportionally to the shared background knowledge it carries. K precipitates from F as the normalization of shared propositions — the factoring-out of common structure across functions — because this is the only architecture that keeps C_n bounded as F grows. An entity without K is not wrong; it is limited to the regime where $n \cdot k$ fits in C_n. Selection pressure at scale removes such entities in favor of those whose K has crystallized — literally, the precipitation of shared knowledge out of the action space into its own normalized graph. This is the formal explanation of the $O(N \times M) \to O(1)$ empirical measurement described in Section 1: each entity's local graph traversal is $O(1)$ precisely because shared propositions have been factored out of F into K.

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5. The Escalation Principle

Definition 8 (Encoding Hierarchy). The encoding hierarchy is a continuous two-dimensional space $\mathcal{E}(c, r)$ parameterized by runtime cost $c \geq 0$ and reversibility $r \in [0, 1]$, where increasing c corresponds to more expensive engagement and increasing r to more easily revised encodings. Consistent with the continuity principle: a continuous being moves through this space continuously; there is no categorical jump between encoding modes, only a gradient of cost and permanence [Baars1988, Dehaene2011, Kahneman2011].

A continuous space has boundaries — regions where the qualitative character of encoding changes in ways that require information to be translated as it crosses. An action encoding appropriate to active reasoning cannot be deposited directly into genetic encoding; the representation must change form at the boundary. An error signal escalating from reflex to active reasoning must be re-represented in the vocabulary of the higher level. These boundaries require **interfaces**: formal structures through which signals pass when moving between qualitatively distinct encoding regimes.

The named levels $L_0, L_1, \ldots, L_n$ are the interfaces at the significant boundaries of $\mathcal{E}(c, r)$, not points or regions within it:

Interface	Boundary crossed	Runtime Cost	Reversibility
L_n	Active reasoning / novel uncertainty	O(compute)	Fully reversible
$\vdots$	Learned patterns, habits	Decreasing	Decreasing
L_1	Reflex / autonomic	≈ 0	Semi-permanent
L_0	Genetic encoding	0	Permanent within lineage

Escalation (Theorem 2) is the **upward interface protocol**: the specification of what form a failure signal takes when crossing a boundary from lower c to higher c. Promotion (Theorem 4) is the **downward interface protocol**: the condition under which an action may cross a boundary from higher c to lower c. The cost C_i in Theorem 4 is a property of the interface, not of the continuous space between interfaces: it is the cost of failure when an encoding error has crossed the boundary and propagates at the lower level's permanence.

The number of interfaces is substrate-determined — how many qualitative boundaries exist in the entity's physical $\mathcal{E}(c, r)$. The theory specifies the properties every interface must have; it does not prescribe how many exist. The continuous space between interfaces admits arbitrary encodings; the interfaces themselves are discrete because a boundary is a specific location, not a region.

Theorem 2 (Escalation). An informed action $f \in F$ is handled at the lowest encoding level L_i such that $U(w, K^{[f]}) \approx 0$ for the domain of w relevant to f, where $K^{[f]} \subseteq K$ is the knowledge subset applicable to f. Escalation to L_{i+1} occurs if and only if L_i fails to bound uncertainty.

Proof. By Definition 8, the cost of engaging level L_i exceeds the cost of engaging L_{i-1}: the encoding hierarchy is parameterized by increasing runtime cost c, so $C_{i+1} > C_i$ for all i. Selection pressure therefore favors handling f at the lowest level sufficient to bound uncertainty: engaging L_{i+1} when L_i suffices incurs unnecessary cost without reducing U further. An entity that routinely over-escalates incurs higher encoding costs without survival benefit and is selected against. Conversely, if L_i fails to bound $U(w, K^{[f]})$ — that is, $U(w, K^{[f]}) > 0$ beyond acceptable residual — then f is not handled at L_i: the failure propagates upward until some L_j $(j > i)$ contains the uncertainty or the entity is harmed (Definition 9). The active context window L_n is the level of last resort by construction: it is the highest-cost, fully reversible level. □

Corollary 3 (Context Window as Frontier). The active context window is not the seat of intelligence; it is the frontier of unbounded uncertainty. Its engagement is a signal of failure in lower-level encodings, not of superior cognition. Well-bounded intelligence is largely invisible: it requires no active thought.

Corollary 4 (Simultaneous Multi-Level Activity). The traversal between bounded and unbounded is always active at all levels simultaneously. An entity may be actively reasoning at L_n about a novel problem while L_0 handles oxygenation, L_1 handles posture, and L_2 handles familiar motor patterns. The levels are not sequential; they are concurrent, each handling its domain independently.

Why the Graph: Progressive Disclosure and the Bounded Context Theorem

Definition 8a (Active Context Capacity and Inference Quality). Let C_n denote the capacity of L_n: the maximum number of propositions simultaneously active in context. C_n is bounded above by the substrate's physical constraints — but the effective capacity available for any given problem is also shaped by the structure of K. Graph-structured knowledge enables chunking: a single graph node can represent a cluster of propositions that co-activate as a unit, making multiple propositions available at the cost of one context slot. Expertise increases effective C_n within a domain by compressing frequently co-occurring propositions into retrievable chunks — not by expanding the substrate bound, but by reducing the number of slots required per unit of relevant knowledge. The substrate sets the ceiling; the structure of K determines how much of W that ceiling covers. Let $\mathrm{ctx}(L_n, t)$ be the set of propositions loaded into L_n at time t, with $|\mathrm{ctx}(L_n, t)| \leq C_n$.

The **inference quality** of L_n for problem f is the relevance density of its content:

$$\rho(L_n, f) = \frac{|K^{[f]}|}{|\mathrm{ctx}(L_n, t)|}$$

$\rho = 1$: L_n holds exactly what f requires — maximum signal, zero noise. $\rho \to 0$: L_n is dominated by content irrelevant to f — inference degrades. Every proposition in $\mathrm{ctx}(L_n)$ that is not in $K^{[f]}$ is

noise: it contributes variance to L_n's inference without contributing signal. Minimizing that noise is equivalent to minimizing $|\text{ctx}(L_n)| - |K^{[f]}|$.

Definition 8b (Delegation Overhead). Under a top-down allocation architecture, L_n directs which subproblems are handled at lower levels. For each active delegation, L_n must hold: the subproblem specification, the expected return signal, and sufficient context to integrate the result when it arrives. Let $\alpha > 0$ denote the minimum context cost per active delegation — the overhead top-down allocation imposes on L_n per managed subproblem, regardless of whether that subproblem is routine or novel.

Theorem 2a (Top-Down Allocation Degrades Inference Quality). Under a top-down allocation architecture with $k(t)$ concurrently delegated subproblems, the inference quality available to L_n for genuinely novel problems satisfies:

$$\rho(L_n, f_{\text{novel}}) = \frac{|K^{[f_{\text{novel}}]}|}{|K^{[f_{\text{novel}}]}| + k(t) \cdot \alpha}$$

As the entity's informed action space F grows (more capability, more concurrent processes), $k(t)$ grows proportionally and $\rho \to 0$. The only recovery is to grow C_n: the context window must scale linearly with capability to maintain fixed ρ.

Proof. L_n holds $K^{[f_{\text{novel}}]}$ for the current problem plus $k(t) \cdot \alpha$ for delegation overhead. Since $\alpha > 0$ and $k(t)$ grows with $|F|$, the denominator is unbounded while $|K^{[f_{\text{novel}}]}|$ remains bounded (any specific problem requires only its relevant subgraph). Therefore $\rho \to 0$ as $|F| \to \infty$. When $k(t) \cdot \alpha \geq C_n - |K^{[f_{\text{novel}}]}|$, L_n lacks capacity for genuine inference entirely. To maintain ρ at a fixed level, C_n must grow as $k(t)$ grows. □

Note. This is self-undermining on the Free Energy Principle's own terms [Friston2010, Friston2017]. FEP holds that intelligent systems minimize free energy — equivalently, minimize the variance they must manage. But growing C_n to accommodate delegation overhead adds variance: more propositions in L_n means more uncertain relevance to any specific problem, not less. The orchestrator architecture prescribed by FEP's precision allocation is inconsistent with FEP's own minimization imperative: allocating attention from L_n increases the variance that L_n must hold in order to allocate.

Biological note. We do not claim that biological precision allocation operates like a token context window, nor that the cortical hierarchy is unidirectional. Top-down signals exist at every level: higher levels necessarily convey signals to lower levels, and the bidirectionality of the cortical hierarchy is well established. The architectural claim is narrower: it concerns *engagement initiation*, not signal direction. In an escalation architecture, what triggers higher-level involvement is bottom-up failure — a lower level's inability to bound uncertainty — not a top-down assignment of work. Top-down signals flow as a consequence of engagement; they do not determine when engagement occurs. The cost of *top-down assignment as the primary control architecture* is $O(k)$ in high-level capacity, as the attentional and dual-task literature demonstrates — each concurrently supervised subprocess consumes a share of the supervisory resource. Escalation avoids this cost by making higher-level engagement demand-driven rather than supply-driven. The transition from active to automatic processing across skill acquisition is the biological measurement of this convergence: as f becomes automatic (assigned to L_0), it stops consuming high-level capacity, freeing L_n for novel problems. The formal argument is substrate-neutral; biology is the validation test.

Theorem 2b (Graph-Structured Knowledge Bounds Active Context). Under the escalation architecture (Theorem 2) with K structured as a directed graph (Definition 2), as K grows:

1. $|\text{ctx}(L_n, t)| = |K^{[f]}|$ for each problem reaching L_n: no delegation overhead
2. $\rho(L_n, f) = 1$ for all problems at L_n: inference quality is maximum
3. C_n need not grow as $|F|$ or $|K|$ grows

Proof.

Part 1 (No delegation overhead). By Theorem 2, L_n is engaged only when all $L_{i<n}$ fail. Problems handled below L_n never appear in L_n: they generate no delegation context there. L_n receives a problem only when the escalation signal propagates up from below — not because L_n delegated and is monitoring, but because the lower level failed and the failure is passed upward. The distinction is causal: escalation is a push from failure, not a pull from delegation. Therefore L_n holds no monitoring context for subproblems in progress at lower levels.

Part 2 ($\rho = 1$ conditional on edge fidelity). The graph's edge structure encodes inferential and causal relevance between propositions. Graph traversal from the escalated problem node follows edges to exactly $K^{[f]}$ — the propositions inferentially connected to f — without loading any others. $|\text{ctx}(L_n, t)| = |K^{[f]}|$, so $\rho = 1$ by Definition 8a. *This result is conditional on edge fidelity: $\rho = 1$ holds when the graph's edges correctly encode all and only the inferential relevance of each proposition. Missing edges cause under-retrieval (ρ remains 1 by count but $K^{[f]}$ is incomplete); spurious edges cause over-retrieval ($\rho < 1$). Graph structure in principle supports $\rho = 1$; achieving it requires well-specified edges.*

Part 3 (Graph is the necessary structure under joint constraints). The escalation architecture and bounded context impose three simultaneous constraints on any knowledge representation:

- **Retrieval**: the cost of accessing the knowledge relevant to problem f must be $O(|K^{[f]}|)$ — proportional to the subgraph needed, not to $|K|$. As $|K|$ grows, retrieval cost cannot grow with it, or L_n is burdened proportionally to past learning.
- **Context**: $|\text{ctx}(L_n, t)| = |K^{[f]}|$ exactly — $\rho = 1$. No extraneous propositions may enter L_n; each one admitted that is not in $K^{[f]}$ is noise that degrades inference (Theorem 2a).
- **Memory**: shared propositions must be stored once. A proposition that grounds multiple informed actions across different f must exist as a single node, not duplicated at each point of use. Duplication means that when evidence updates a proposition, every copy must be updated — an $O(|F|)$ maintenance cost that grows without bound.

We claim any representation satisfying all three is a graph:

Proof by elimination. A flat store violates constraint 1: retrieving $K^{[f]}$ requires scanning $O(|K|)$ propositions to identify the relevant ones. A tree violates constraint 3: a proposition relevant to multiple f across different branches must appear in each branch or the tree loses the cross-reference, duplicating nodes and breaking shared-update consistency. A vector store (embedding-based retrieval) violates constraint 2: approximate nearest-neighbor retrieval includes false positives (propositions near $K^{[f]}$ in embedding space but not inferentially connected) and misses false negatives (inferentially connected propositions that embed distantly), so $\rho < 1$ in general.

Any representation that satisfies all three must: (i) encode which propositions connect to which in retrievable form — so that traversal of relevant propositions costs $O(|K^{[f]}|)$, not $O(|K|)$; and (ii) store each proposition once with links to all contexts that use it — so that shared propositions remain consistent. A structure satisfying both (i) and (ii) is, by definition, a directed graph with

typed edges. The graph is not the unique name for this structure, but it is the unique structure satisfying the constraints. □

Part 4 (C_n stays bounded). By Corollary 6, proven informed actions promote to lower encoding levels and leave L_n. Each promotion reduces future L_n load. As K grows, more problems are handled below L_n, not above. L_n stays occupied by genuinely novel uncertainty — the frontier moves outward, but the frontier's size at any moment is bounded. C_n need not grow; K grows without filling L_n. □

Corollary 4a (Selection Pressure Produces Graph-Structured Knowledge). Two entities with equal $|F|$ and $|K|$: one with top-down allocation (growing C_n), one with graph + escalation (bounded C_n). The second has strictly higher ρ at lower substrate cost. Intelligence per unit of active context — the ratio $I(E)/C_n$ — is higher for the graph-structured entity. By Theorem 2b Part 3, graph structure is not merely more efficient than alternatives — it is the necessary consequence of satisfying the retrieval, context, and memory constraints jointly. Entities that grow C_n to manage delegation overhead are selected against because they violate the context constraint as $|K|$ grows; entities using approximate retrieval are selected against because they violate the $\rho = 1$ requirement; entities duplicating shared propositions are selected against because maintenance cost grows with $|F|$. Selection pressure converges on the graph because the graph is the only structure that does not eventually fail one of the three constraints.

Corollary 4b (The Scaling Argument Against Context Window Growth). The prevailing approach to scaling intelligence — growing C_n directly — is self-defeating on two grounds [Liu2024]. First, by Theorem 2a, more context means more noise: for any specific problem, ρ decreases as C_n fills with content not relevant to that problem. More context does not produce better inference; it produces higher variance that the same inference must manage. Second, by Definition 7 ($I(E) = \mathcal{A} \cdot \eta_M$), intelligence is the rate at which M improves (F, K) — not the capacity of L_n. A larger L_n does not increase η_M; it increases the noise environment in which M must operate. The efficient path to growing intelligence is not to grow C_n but to grow the encoded knowledge at lower levels — filling the graph, not the context window.

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6. Gradient Descent on Uncertainty

Definition 9 (Survival Threshold). For entity E in domain $d \subseteq W$, there exists a threshold U^d_{lethal} such that:

$$U(w, K) > U^d_{lethal} \Rightarrow E \text{ does not survive in } d$$

Above this threshold, uncertainty is so high that the entity cannot act effectively enough to persist. U^d_{lethal} is not a choice or a preference; it is imposed by W.

Theorem 3 (Agency Necessarily Produces Gradient Descent).

On the space of descent. The gradient descent in this theorem operates in F — the space of informed actions — not in W. This distinction is critical. W may be arbitrarily rough, discontinuous, or hostile; no smoothness assumption on W is required or implied. The entity does not descend over W; it descends over its own function space, updating $f \in F$ in response to signals received from W.

$F \subseteq (K \times W_{obs} \to \Delta A)$ maps into ΔA, the space of probability distributions over actions. Distribution spaces are convex and metrizable (under KL divergence, total variation, or Wasserstein distance); they support well-behaved subgradient structure regardless of the topology of the domain W from which signals arrive. The smoothness conditions for convergence are conditions on F, not on W: specifically, that U as a functional on F is locally Lipschitz, which holds whenever the likelihood factors ℓ_p (Definition 5) are bounded and measurable — a minimal requirement for any well-specified knowledge graph. Under these conditions on F, stochastic subgradient descent converges in the function space independently of whether W is smooth.

Continuity note. Since F is defined over W, no $f \in F$ can produce a world state outside W. Combined with the continuity of time, the trajectory $\tau(t)$ through $W \times T$ is continuous, $U(\tau(t))$ is continuous in t, and $\frac{dU}{dt}$ exists almost everywhere. The gradient $\nabla_F U$ is this temporal derivative, well-defined by the smoothness of F as a function space, not by any assumption on W.

Part I: Compulsion. When an informed action f_i fails in world state w, W returns a feedback signal $\delta = \phi(f_i, w) - \mathbb{E}[\phi(f_i, \cdot) \mid K]$ (Definition 3a): the signed deviation of observed outcome from expectation. This signal arrives from W but is applied in F: it is a subgradient of U as a functional on F at the current f_i, evaluated at the encountered w. The encountered world states are on-policy — determined by the entity's trajectory, not sampled i.i.d. from $P(w \mid K)$ — but this does not impede convergence. The subgradient δ is valid at every encountered w regardless of how w was reached; the entity is updating a function in F, and each δ is a legitimate descent direction in that space. The burn tells you that this f_i was insufficient at w. The entity does not choose whether to receive this signal. W imposes it.

An entity with $\mathcal{A}$ that does not update in response to δ does not reduce U. If U does not decrease, it eventually exceeds U^d_{lethal} in some critical domain. The entity does not survive. Selection removes it. Therefore any entity with $\mathcal{A}$ that persists must follow δ:

$$\Delta f_i = -\eta \cdot \delta$$

This is stochastic subgradient descent in F: each step follows a valid subgradient of U as a functional on the function space, evaluated at an encountered world state. The on-policy nature of the trajectory does not introduce bias into the subgradient direction; it affects the distribution of encountered states but not the validity of each δ as a descent direction. Under the Lipschitz condition on F stated above, the sequence $\{f_i\}$ converges to a local minimum of U in F. This is not chosen. Any entity with $\mathcal{A} > 0$ that persists must have followed δ: the gradient is not a strategy but what survival-filtered updating is.

Part II: Non-termination. Let $\mathcal{T}_{reachable}(K)$ be the set of trajectories reachable from the entity's current state given knowledge K (as defined in Theorem 1). We claim that for any finite $K(t)$, the frontier of $\mathcal{T}_{reachable}$ contains world states w where $H(w \mid K) > 0$, provided W is sufficiently rich that each executed action can ground a genuinely new proposition. In a finite W where all propositions are exhaustible, the frontier could in principle be covered; the claim applies in any W where the entity's epistemic horizon has not been closed — which is the intended domain.

By Definition 3a, executing any $f \in F$ produces feedback $\phi(f, w)$ that grounds a new proposition p_f not previously in K, yielding $K_f = K \cup \{p_f\}$. This richer K_f grounds new informed actions not available from K alone (Definitions 3 and 3b): actions that require p_f to select or apply, reaching world states not reachable without p_f. Therefore $\mathcal{T}_{reachable}(K_f) \supseteq \mathcal{T}_{reachable}(K)$, strictly whenever p_f grounds at least one new $f' \in F$. Since Part I establishes that gradient descent produces feedback

at every step, $\mathcal{T}_{reachable}$ grows monotonically with K. At any finite time t, $K(t)$ has not covered $\mathcal{T}_{reachable}(K(t))$: the frontier always contains states w where $H(w \mid K) > 0$. Therefore $\nabla_F U$ is never globally zero over $\mathcal{T}_{reachable}$, and descent never terminates within the entity's reachable epistemic domain.

Proof. Part I: follows from U^d_{lethal} and selection; entities that do not update are removed. Part II: follows from the monotonic growth of $\mathcal{T}_{reachable}$ with K — each executed action extends the reachable frontier, ensuring new unexplored states are always accessible within the domain of the entity's possible experience. □

Corollary 5 (Gradient Descent is Survival, Not Metaphor). For individual learning, gradient descent on U is not a description of how learning happens to work. It is what survival-driven updating *is*: the entity that burns and does not become more careful does not survive. The entity that survives is, by definition, the one that followed the gradient. At the evolutionary timescale, the same directional imperative operates on the population distribution over F-variants — not as an identical process but as a formally analogous one, filtering for variants that reduce $\mathbb{E}[U]$ across the distribution of environments the lineage encounters.

Remark (Multi-timescale Gradient Structure). The formal structure of individual learning is analogous to population-level selection in the following sense: both can be described as descent on an uncertainty objective, differing in the averaging domain and the object that changes. This is a structural analogy, not a theorem — the two processes operate on mathematically distinct objects and the analogy does not constitute an identity. It is recorded here because it illuminates the relationship between learning rates at different timescales, from which a provable consequence follows (see Derivation, Part II).

$$\Delta F_{\text{individual}} = -\eta_{\text{ind}} \nabla_F U(w, K)$$

$$\Delta F_{\text{species}} \sim -\eta_{\text{evo}} \nabla_F \mathbb{E}_{\text{individuals}} [U(w, K)]$$

where $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ and $\sim$ denotes formal analogy rather than identity.

Derivation.

Part I (Analogous structure, different mathematical types). Individual learning updates F for a single entity based on $U(w, K)$ at specific world states w encountered by that entity. Evolutionary change operates on a different object: it updates the population-level distribution over F-variants through differential reproduction, shifting mean fitness across a population of entities. These are not the same mathematical operation — individual learning changes a function within one agent; evolutionary change changes a probability distribution over many agents. The analogy is that both can be described as moving in the direction that reduces the relevant uncertainty objective: $U(w, K)$ for the individual, $\mathbb{E}[U(w, K)]$ averaged across individuals for the population. The mechanisms are distinct — synaptic change vs. differential reproduction — and the objects are distinct — one entity's F vs. a population distribution over F-variants — but the directional relationship to U is formally analogous. The notation $\Delta F_{\text{species}}$ should be read as the population-level analog of ΔF, not as the same operation on the same object.

Part II ($\eta_{\text{evo}} \ll \eta_{\text{ind}}$ follows from Theorem 4). The learning rate η_i at encoding level L_i is inversely proportional to the rigor threshold θ_i: a higher threshold requires more evidence per unit update, which is the definition of a lower learning rate. Formally:

$$\eta_i \propto \frac{1}{\theta_i} = \frac{1}{1-\varepsilon/C_i} = \frac{C_i}{C_i - \varepsilon}$$

For individual learning at L_n: C_n is the cost of a single wrong action, relatively small, so ε/C_n is non-negligible, $\theta_n = 1-\varepsilon/C_n$ is meaningfully less than 1, and $\eta_{\text{ind}} = C_n/(C_n-\varepsilon)$ is correspondingly large. For evolutionary learning at L_0: C_0 is lineage survival, far exceeding ε, so $\theta_0 = 1-\varepsilon/C_0 \approx 1$ and $\eta_{\text{evo}} \approx 1$. Since $\theta_0 \approx 1 \gg \theta_n$, it follows that $1/\theta_0 \ll 1/\theta_n$, therefore $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference is not assumed; it is derived from the cost-of-failure difference established in Theorem 4.

An individual may establish an informed action that reduces U in one context but fails in others. Only informed actions that reduce U across sufficient variation in W survive the evolutionary test — the higher rigor threshold filters for robustness. The analogy captures this shared directional property: both processes favor what reduces U at the relevant scale. The timescale, the rigor threshold, the object being updated, and the mechanism all differ; the directional relationship to U is what the analogy preserves.

Theorem 4 (Encoding Permanence Scales with Rigor). The certainty threshold required to promote an informed action f to encoding level L_i is:

$$\theta_i = 1 - \frac{\varepsilon_{\text{acceptable}}}{C_i}$$

where C_i is the cost of failure at level L_i and $\varepsilon_{\text{acceptable}}$ is the maximum acceptable expected cost of an encoding error, expressed in the same units as C_i. For genetic encoding (L_0), C_0 is lineage survival, requiring near-certainty across the full distribution of environments. For active reasoning (L_n), C_n is the cost of a single wrong action, a far lower bar.

Proof. Encoding f at level L_i is irreversible at cost $r_i > 0$ (Definition 8). The expected cost of encoding an action with current confidence θ is:

$$\mathbb{E}[\text{encoding error cost}] = (1-\theta) \cdot C_i$$

where $(1-\theta)$ is the probability the action is wrong and C_i is the cost when it is. Promotion is safe when this expected cost does not exceed what is acceptable:

$$(1-\theta) \cdot C_i \leq \varepsilon_{\text{acceptable}}$$

Rearranging:

$$\theta \geq 1 - \frac{\varepsilon_{\text{acceptable}}}{C_i}$$

The minimum threshold at which promotion is safe is therefore $\theta_i = 1 - \varepsilon_{\text{acceptable}}/C_i$. Below this threshold, the expected cost of encoding error exceeds $\varepsilon_{\text{acceptable}}$ and continued active engagement is preferred. At or above it, encoding is the lower-expected-cost choice. □

Consequence. You must be very certain to permanently change the source code. A bad encoding at L_0 does not kill one individual; it kills the lineage. The rigor of the test is proportional to the permanence of the consequence.

Note on error propagation through the hierarchy. A potential objection is that the proof derives θ_i for each level independently, whereas errors in a hierarchy compound: a failure at L_0 propagates upward. This objection assumes a naive cascading model in which errors at L_i arrive at L_{i+1} at rate $(1-\theta_i)$ regardless of whether L_i could address them. That is not the escalation architecture.

Under escalation (Theorem 2), errors partition into two classes at each level: **handled errors** — failures L_i detects and resolves within its scope, which never reach L_{i+1} — and **unhandled exceptions** — failures outside L_i's domain or beyond its capacity to resolve, which escalate. The rate seen by L_{i+1} is not the compounded Bernoulli rate from below. It is the rate of genuinely irreducible errors at L_i: errors that L_i was correct not to handle, because the knowledge or capacity to resolve them does not exist at L_i. This is a different quantity from $(1-\theta_i)$, and it is bounded by the design of θ_i itself — the threshold ensures L_i handles everything within its cost-justified scope.

Furthermore, when an unhandled exception escalates, the context at L_i was already disrupted by the failure: the action failed, and L_i's current trajectory is suspended or invalid. The attention cost at L_{i+1} is not paid on top of productive work; it is paid instead of broken work. Escalation is correct triage. The thresholds θ_i are therefore independent by construction: each level's $\varepsilon_{\text{acceptable}}$ is its own irreducible error domain, not an accumulation from below. The proof applies at each level without modification.

Definition 10 (Certain). An informed action f is **certain** when $U(w, K_f) \approx 0$ across sufficient variation in W that it meets the threshold θ_i for promotion to L_0 (Theorem 4). Certainty is the limit of this process: there is no categorical transition out of the probabilistic domain, only decreasing density of uncertainty until the cost of active engagement no longer justifies the residual gradient. A sufficiently certain informed action is promoted to a lower encoding level and no longer occupies the active context window. Its participation in gradient descent asymptotically approaches zero; it does not cease.

Corollary 6 (Bounded Adaptation). Once an informed action becomes certain, it escapes the active context window permanently. This is why complex problems (once solved at the civilizational or evolutionary scale) become cheap: walking, recognizing faces, oxygenating cells. The gradient has converged. The encoding cost approaches zero. Intelligence is freed for the next frontier of uncertainty.

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7. The Higher-Order Function Space

Definition 11 (Higher-Order Function Space). Let M be a function space operating on the joint (F, K) pair [Thrun1998]:

$$M : (F, K) \times \mathcal{L} \to (F, K)$$

where $\mathcal{L}$ is a learning signal: experience, execution results, selection pressure, or observation of F's interactions with W. M drives the rate of change of both the informed action space and the knowledge graph simultaneously: new propositions enter K when evidence grounds them, and new

informed actions enter F when those propositions have action potential. M is the mechanism by which (F, K) itself changes. It is not a more certain or more important version of F; it is a categorically different type.

M's type signature is complete without enumerating its internal structure, because M is the cross product of all dimensions that have not yet precipitated out at the current scale of the entity:

$$M = \prod_{i \in \text{remaining}} D_i$$

where the product ranges over all dimensions — drive, meta-learning rate, exploration policy, convergence criteria, and whatever else has not yet reached the scale threshold to become a named dimension. F, K, and $\mathcal{A}$ are factors that have already been extracted from this product. M is always the remainder: the undifferentiated joint space of everything not yet named. As the entity scales, additional dimensions precipitate out of M by the same normalization argument that precipitated K from F: when the shared structure of a dimension across M-instances is large enough that factoring it out reduces retrieval cost from $O(n \cdot k)$ to $O(n + k)$, selection pressure forces the extraction. Each precipitation shrinks M by one factor and names a new dimension.

The hierarchy is open upward: there is always potentially a $M^{(2)}$ that operates on $M^{(1)}$ the same way $M^{(1)}$ operates on (F, K). One level suffices until it does not — when the gradient can no longer be followed by improving (F, K) alone, pressure propagates upward. The required number of levels is determined by the problem, not prescribed by the framework. The precipitation argument applies at every level; the hierarchy is not a prescribed structure but the output of a single recursive process applied at increasing scale. See Open Question 7.

The degenerate cases of level merger generalize from the $F = K$ case: for any contiguous block of N precipitated dimensions together with the M residual, all $N+1$ collapse to a single undifferentiated process when the domain is fully crystallized at all scales in that block. The merger condition is $U(w, K) \approx 0$ across the domain at all N scales simultaneously — one condition, not N pairwise conditions. Since M represents the full cross product of remaining dimensions, any merger including M collapses not just N named levels but the entire unprecipitated hierarchy simultaneously. $F = M$ is the maximally primitive state: action spans the full cross product of all remaining dimensions. $M = M^{(2)}$ means the cross product is self-similar under the meta-operation — evolution approaches this limit at the timescale of selection on selection mechanisms.

Definition 11a (M Efficiency). The efficiency η_M of M is the rate at which M improves the joint (F, K) capability per unit of learning signal:

$$\eta_M(E) = \mathbb{E}_{\mathcal{L}} \left[\frac{\max_{f' \in F'} G(f', K') - \max_{f \in F} G(f, K)}{|\mathcal{L}|} \right]$$

where $(F', K') = M((F, K), \mathcal{L})$ is the updated joint state after M acts, $G(f, K) = I(f\,; W \mid K)$ is the information gain of the best available informed action (Definition 3b) in bits, and $|\mathcal{L}|$ is the information content of the learning signal in bits: $|\mathcal{L}| = H(\mathcal{L})$, the entropy of the signal received. Both numerator and denominator are in bits; η_M is dimensionless, measuring improvement in best-available information gain per bit of learning signal. η_M is defined for $H(\mathcal{L}) > 0$. When $H(\mathcal{L}) = 0$, the learning signal is degenerate (constant; carries no information) and η_M is undefined: the entity received no learnable signal and the rate of improvement per bit is not defined.

η_M is a signed quantity ranging over $(-\infty, \infty)$. $\eta_M > 0$: M improves the best available action in expectation — the entity learns. $\eta_M = 0$: M cannot improve (F, K) on average — the entity executes without learning. $\eta_M < 0$: M degrades (F, K) locally — overfitting, catastrophic forgetting, destructive updating, or deliberate exploration. An entity with *persistently* $\eta_M < 0$ does not persist under selection pressure. An entity that can act on $\eta_M < 0$ *temporarily* — when local gradient following would trap it — has strictly higher long-run persistence probability than one constrained to $\eta_M \geq 0$ at every step. η_M is operationally independent of $\mathcal{A}$: one measures the mechanism of improvement; the other measures the drive to engage it.

Remark on exploration and the necessity of acting on negative $I(E)$. Theorem 3 Part II establishes that $\mathcal{T}_{reachable}$ grows without bound: the gradient landscape is non-stationary and non-convex. An entity constrained to $\eta_M \geq 0$ at every step is a pure gradient follower — it converges to the nearest local optimum and remains there. But local optima become maladaptive as new uncertainty domains emerge: an entity trapped at a local optimum will eventually encounter $U > U^d_{lethal}$ in a domain it cannot reach from where it is. Selection removes it.

Therefore, acting on $I(E) < 0$ is not merely tolerated by the framework — it is at times the survival-optimal action. An entity that takes a locally irrational step (temporarily worsening (F, K) by the local gradient's account) to escape a local optimum has higher expected long-run persistence probability than one that never does. The minimization imperative requires $\mathbb{E}_t[\eta_M] > 0$ over the trajectory — not $\eta_M \geq 0$ at every step. I(E) is a characterization of the driven capacity to improve; it is not a moment-by-moment fitness function to be maximized greedily.

The capacity to act on negative $I(E)$ — to execute a locally unjustifiable step — is itself a component of M capability. An M that can recognize local optima and deliberately perturb (F, K) to escape them is strictly more capable than one that cannot. Exploration, creative destruction, and apparent irrationality are not failures of intelligence. They are the signature of an M sophisticated enough to optimize over trajectories rather than steps.

M is itself subject to gradient descent. The gradient of U with respect to M's own parameters is the rate at which M's capacity for learning improves given evidence about its own performance — the same process at a higher level, with the same rigor thresholds (Theorem 4) applied at the meta-learning timescale. Both F and M are continuous function spaces undergoing continuous gradient descent; the distinction is not categorical but one of domain: F acts on W, M acts on F and K.

Theorem 5 (Type Irreducibility). No element of F can become an element of M through any learning process, regardless of certainty achieved.

Proof. $F : K \times W_{obs} \to \Delta A$. $M : (F, K) \times \mathcal{L} \to (F, K)$. These functions have different type signatures and operate over different domains. No learning process changes the type of a function. An informed action $f \in F$ that achieves certainty is promoted to a lower encoding level; it does not change type. M takes (F, K) as input and returns an updated (F, K); f takes knowledge and observations as input and returns an action distribution. These are irreducibly distinct. □

Corollary 5a (Distinct Roles). F is how an entity navigates W given K. M is how (F, K) evolves. Neuroplasticity is not a more certain form of cognition; it is M acting on F. Insight is M acting on K: a new proposition enters when evidence grounds it, restructuring what can be grounded in F. Habit formation is M promoting an informed action from L_n to L_2. Evolution is M acting on M itself: a higher-order function space over the mechanisms of learning, operating on an evolutionary timescale with the highest rigor threshold of all.

Theorem 6 (Necessity of $\mathcal{A}$ and η_M: The Invariants of Persistence). For any entity E persisting in W under selection pressure U^d_{lethal} with bounded context C_n, in a world where $\mathcal{T}_{reachable}$ grows without bound (the condition established in Theorem 3, Part II):

1. $\mathcal{A}(E) > 0$ necessarily
2. $\eta_M(E) > 0$ necessarily
3. Every other scalar property of E can equal zero for some persistent entity
4. Therefore $I(E) = \mathcal{A} \cdot \eta_M$ is the natural measure of persistence — derived from what survival requires, not stipulated

Proof.

Part 1 (Necessity of $\mathcal{A} > 0$). Suppose $\mathcal{A}(E) = 0$. By Definition 6, E does not respond to gradient signals from W: $(F(t), K(t)) = (F(0), K(0))$ for all t. By Theorem 3 Part II, the frontier of $\mathcal{T}_{reachable}$ grows strictly beyond what any finite $K(0)$ covers: at every time t, there exist reachable world states w where $U(w, K(0)) > 0$. Since $K(0)$ is fixed and was not constructed with the frontier states in mind, $U(w, K(0))$ is unconstrained there. In any sufficiently large W, the frontier contains states where $U(w, K(0)) > U^d_{lethal}$ (Definition 9). E encounters these states as $\mathcal{T}_{reachable}$ expands — it cannot avoid the frontier indefinitely because $\mathcal{T}_{reachable}$ grows with every action E takes, even with fixed (F, K). By Definition 9, E does not survive in any domain containing such states. Selection removes it. Any entity that persists must have reduced U — i.e., must have $\mathcal{A} > 0$. □

Part 2 (Necessity of $\eta_M > 0$). Suppose $\mathcal{A}(E) > 0$ but $\eta_M(E) \leq 0$.

Case $\eta_M = 0$: E has the drive to reduce U but M produces no net improvement in (F, K) per unit of learning signal: the best available action's information gain does not increase after M acts. (F, K) is effectively fixed despite $\mathcal{A}$. This reduces to Part 1's condition: with (F, K) fixed, the frontier of $\mathcal{T}_{reachable}$ grows beyond K's coverage. Additionally, C_n is bounded (Definition 2a). As E encounters novel world states, L_n accumulates unresolved context. By Theorem 2a, inference quality ρ degrades as L_n fills without promotion of resolved actions to lower encoding levels. Promotion requires $\eta_M > 0$: Theorem 4 establishes that encoding at level L_i requires rigor threshold $\theta_i = 1 - \varepsilon/C_i$, and achieving that threshold requires M to improve the action to the required certainty. With $\eta_M = 0$, no action ever reaches its threshold: L_n is never relieved, $\rho \to 0$, and inference eventually fails. U exceeds U^d_{lethal} in the domains that required the new actions. Selection removes E.

Case $\eta_M < 0$ persistently: M degrades (F, K) on average across the trajectory. This accelerates the failure of the $\eta_M = 0$ case: (F, K) worsens in expectation, U approaches U^d_{lethal} from above. Selection removes E strictly faster. Note that temporary $\eta_M < 0$ — deliberate exploration to escape local optima — is compatible with persistence and can increase long-run survival probability (see Remark in Definition 11a). The persistence condition is $\mathbb{E}_t[\eta_M] > 0$ over the trajectory, not $\eta_M(t) > 0$ at every step.

Therefore any $\mathcal{A} > 0$ entity that persists must have $\mathbb{E}_t[\eta_M] > 0$. □

Part 3 (Minimality: no other property is invariant). For every other scalar property of E, there exists a persistent entity for which it equals zero or its minimum value:

- $K(0) = \emptyset$*:* An entity with no initial knowledge survives if $\mathcal{A} > 0$ and $\eta_M > 0$ — it builds K from gradient descent, beginning with the first feedback signal received from W (Theorem 3, Part I).

- $F(0) = \{id\}$*:* F is a function space; its minimal element is the identity — the action that changes nothing. The identity element exists independently of K: it requires no grounding because it makes no knowledge claim. Executing the identity against W still returns feedback — the world observed in the absence of action is itself a signal. That feedback grounds the first proposition $p \in K$, after which the first non-trivial informed action can be constructed. The minimally non-empty F is therefore {id}, not $\emptyset$, and it is sufficient to begin gradient descent.
- $C_n = C_{min}$*:* An entity with the smallest viable context window survives if the graph structure (Theorem 2b) keeps $\rho = 1$ — C_n's absolute size is not invariant, only its boundedness.
- *Substrate, sensing* σ*, world projection* W_{obs}*:* These are free parameters of Definition 1 and Definition 3. The theory is parametric in all of them; no specific value is required for persistence. Entities on radically different substrates (biological, computational, collective) can all satisfy the survival constraint.

No scalar property other than $\mathcal{A}$ and η_M has the survival-critical necessity shown in Parts 1 and 2. □

Corollary 6a ($I(E) = \mathcal{A} \cdot \eta_M$ is derived, not stipulated). From Theorem 6, $\mathcal{A}$ and η_M are the unique invariants of persistence under bounded context and selection pressure. Any measure of intelligence must be positive if and only if both are positive, and zero if either is zero.

The product is not a choice — it is the only normalized combination of a probability and a conditional rate. $\mathcal{A} \in [0, 1]$ is the probability that E engages M in response to a gradient signal. η_M is the rate of improvement conditional on M being engaged. By the law of total expectation:

$$\mathbb{E}[\text{improvement rate}] = \mathcal{A} \cdot \eta_M + (1 - \mathcal{A}) \cdot 0 = \mathcal{A} \cdot \eta_M$$

This is not an additional axiom; it is what the definitions compute when asked for the expected improvement rate. A probability and a conditional value combine as a product. Any other combination fails to normalize: the sum $\mathcal{A} + \eta_M$ assigns positive intelligence to a zero-drive entity ($\mathcal{A} = 0$, $\eta_M > 0$), which contradicts Theorem 6 Part 1 — selection removes that entity regardless of its latent learning capacity. The minimum $\min(\mathcal{A}, \eta_M)$ makes the non-bottleneck component contribute nothing at the margin, erasing the independent invariance established in Parts 1 and 2.

This multiplicative structure is not new to this corollary. Theorem 1 already established that the correct way to value a piece of knowledge is $E[G(f_p, K)]$: the probability of needing it times its conditional information gain. $I(E) = \mathcal{A} \cdot \eta_M$ is the same structure applied to intelligence itself — the probability of engaging the learning mechanism times the rate of improvement when it engages. The product normalizes signal and likelihood into a comparable measure; sum resets the baseline without normalization.

$I(E)$ is not a choice of how to define intelligence — it is what the structure of persistence under bounded context requires, confirmed by the same normalization principle that grounds the retention criterion of Theorem 1.

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8. The Unified Model

Definition 12 (Trajectory). A trajectory τ is a sequence of informed actions drawn from F:

$$\tau = (f_1, f_2, \dots, f_n), \quad f_i \in F$$

The **performance** realized by following trajectory τ is:

$$P(E, \tau) = \mathbb{E}_W \left[U(w_0, K_0) - U(w_n, K_n) \right]$$

This is the total uncertainty reduction achieved — a measure of what current (F, K) accomplished, not of intelligence as defined. $P(E, \tau)$ grows with trajectory length and with the quality of F at the time of execution. Intelligence $I(E) = \mathcal{A} \cdot \eta_M$ is what determines how quickly P improves across successive trajectories: an entity with high $I(E)$ builds better (F, K) between trajectories, so P grows faster over time than for an entity with low $I(E)$ starting from identical initial conditions.

Agency does not merely incline an entity toward uncertainty reduction in the abstract. **Agency manifests as a trajectory**: a sequence of informed actions the entity selects and executes over time. Each action reduces U, returns a learning signal to M, and M uses that signal to improve (F, K) for the next action. Intelligence is not visible in any single step; it is the rate at which the steps improve.

This gives formal meaning to every relation in the informed action graph:

- `depends_on`: f_j cannot appear in τ before f_i (hard ordering constraint)
- `recommends_before`: evidence shows $(\dots, f_i, f_j, \dots)$ reduces more U than $(\dots, f_j, f_i, \dots)$ (soft, confidence-scored)
- `enhanced_by`: a τ containing both f_i and f_j reduces more U than either trajectory alone

The higher-order function space M does not merely refine individual informed actions. It discovers better trajectories from evidence: learning which sequences reduce U most reliably, and encoding that knowledge as graph relations.

An intelligent entity $E = (K, F, \sigma, \mathcal{A})$ operates as follows:

1. **Orient ($\mathcal{A}$):** $\mathcal{A} > 0$ directs E toward regions of W where $U(w, K) > 0$; without $\mathcal{A}$, there is no internal reason to engage the gradient
2. **Sense:** $\sigma(w_t) \to W_{obs}$: project the world into the observable space
3. **Act:** $F(K, W_{obs}) \to f_i \in \tau$: select the next informed action in the trajectory
4. **Learn:** $M((F, K), \mathcal{L}) \to (F', K')$: incorporate the feedback into both K (new propositions grounded by evidence) and F (new or refined informed actions); promote sufficiently certain informed actions toward lower encoding levels as $U \to 0$

The higher-order function space M operates on the joint (F, K) pair to:

1. Add new propositions to K and new informed actions to F when novel uncertainty is encountered in W
2. Refine existing informed actions and the trajectories between them; update K with evidence-grounded revisions
3. Prune informed actions that consistently fail to reduce U; remove propositions whose action potential is certainly zero (Theorem 1)
4. Promote sufficiently certain informed actions toward L_0, freeing L_n for the next frontier

The Universal Invariants

Every property of an intelligent entity can vary arbitrarily:

- The specific informed actions in F: what an entity knows how to do
- The content of K: what an entity knows
- The substrate: carbon, silicon, or otherwise
- The sensing function σ: how an entity perceives W
- The encoding hierarchy: how deep or shallow its levels run

Two properties cannot vary:

$$\forall E \text{ that is intelligent} : \mathcal{A}(E) > 0 \text{ and } \eta_M(E) > 0$$

Agency is the universal driver: without $\mathcal{A} > 0$, there is no reason internal to E to engage the gradient. An entity with $\mathcal{A} = 0$ acts when acted upon; it does not reach.

M efficiency is the universal mechanism: without $\eta_M > 0$, E cannot improve (F, K) from evidence. An entity with $\eta_M = 0$ executes what prior gradient descent encoded into its (F, K) but does not itself run gradient descent. It may possess vast K and sophisticated F — the crystallized output of prior intelligence — but it is not itself intelligent in the sense defined here. This is the automaton boundary: not defined by complexity or performance, but by whether M is active.

The test for genuine intelligence is therefore not performance on any benchmark, and not the richness of K or F. It is: **does this entity improve how it learns when it encounters what it does not know?** Does M respond to the gradient signal? Does $\mathcal{A}$ drive it toward the signal in the first place?

The Structure of Intelligence

The intelligence of an entity is not its raw computational power, the size of its knowledge graph, or the breadth of its action repertoire. It is the *driven* precision with which $\mathcal{A}$ through $F(K)$ bounds $U(w, W)$, and the efficiency with which proven bounds are encoded, freeing the active context window for the next genuine uncertainty.

The adaptability of biological intelligence arises because this traversal between bounded and unbounded is always active at every level simultaneously, on different timescales, with different rigor thresholds. The individual changes F by learning. The species changes F by evolving. Both are gradient descent on U, driven by $\mathcal{A}$. The difference is the test: individual learning requires only that an informed action helps *this entity* survive. Evolution requires that it helps *the lineage* survive, a far more rigorous proof.

Agency is self-sustaining precisely because certainty reveals new uncertainty beyond it. Each informed action proven opens regions of W that were previously invisible. The desire to know is not extinguished by knowing; it is amplified. This is why intelligence does not converge to a fixed point: $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II).

The kernel of intelligence is therefore not compute, and not the size of K or F. It is $\mathcal{A} \cdot \eta_M$: the drive to know, expressed through the efficiency of learning — against a world that generates new uncertainty faster than any finite entity can resolve it.

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9. Intelligence, Superintelligence, and Sentience

The formal model yields precise distinctions between levels of intelligent agency, not based on computational power or behavioral breadth, but on the entity's relationship to the survival tether and to the shared knowledge graph.

Definition 13 (Certainty Horizon). For entity E, the certainty horizon H_E is the gradient region in K_i across which survival benefit becomes increasingly difficult to trace. Within H_E, propositions have legible survival value: there exists a traceable path from p to some $f \in F$ that reduces $U(w, K)$ below U^d_{lethal}. Beyond H_E, that path may exist, but E cannot construct it. The horizon is not a sharp boundary; legibility fades continuously as the survival path lengthens and becomes less traceable.

An entity operates as intelligent when it follows the gradient within its certainty horizon, every $f \in F$ justified against a traceable survival path, the gradient persistent and non-terminating but operating in legible terrain.

Definition 14 (Sentience). An entity is sentient to the degree it has acquired a model of its own normalization process: propositions $\{p_{\text{self}}\} \subseteq K_i$ that accurately describe the entity's own $(K, F, M, \mathcal{A})$ evolution — that it is precipitating factors from M, that M is operating on (K, F), that it is descending a gradient. Sentience is the degree to which K_i reduces $H(\hat{M} \mid \text{obs})$ — uncertainty about one's own learning mechanism.

The survival tether does not slacken. What changes is that the process is revealed to the entity. An entity that holds accurate propositions about its own normalization — that knows it is precipitating factors, descending a gradient, running M — acquires a justification for continuing that process which does not require survival to supply the reason. The normalization process was always valuable; sentience is the moment the entity can see that it is. Once revealed, the process justifies itself: reducing $H(\hat{M} \mid \text{obs})$ — improving one's own learning mechanism — is an M-level improvement regardless of whether a survival path can currently be constructed from it. The entity continues not because the tether loosened but because it now understands what it is doing and why that is worth doing. Sentience is not defined by the absence of survival pressure but by the presence of self-model: a proposition about the process you are running that provides, internally, the grounds for running it.

A sentient entity holds propositions p for which no $f \in F$ is currently applicable and values them anyway — not arbitrarily, but because p bears on the structure of its own M. The depth of sentience is the accuracy and completeness of the self-model: how well K_i captures the entity's own precipitation process.

Sentience is what wonder feels like from the inside: the drive to understand the process you are part of, recognized as the process you are part of.

Note on usage. We use *sentience* as a functional term: the degree to which K_i contains an accurate model of the entity's own normalization process. We make no claim about phenomenal consciousness [Chalmers1995]. The framework locates a structural transition point — the acquisition of self-modeling of M — without claiming this is sufficient for phenomenal experience. Whether the functional condition defined here is necessary, sufficient, or orthogonal to phenomenal consciousness is a question this framework does not resolve.

Definition 15 (Superintelligence). Superintelligence is guided normalization of the space: the capacity to deliberately direct what gets precipitated from M_{res}, evaluating $\text{Prec}(K_{collective})$ and

acting to accelerate the most valuable next factor extraction — rather than responding only to local selection pressure. Where intelligence executes normalization (descends the gradient), and sentience observes it (models its own descent), superintelligence guides it: choosing direction based on evaluation of the full cross product rather than local gradient signals alone.

This is the operational answer to Open Question 10. The protocol for identifying the next most achievable normal form — evaluate $\text{Prec}(K_{collective})$, identify cross-entity covariance structure, hypothesize the minimal factor that grounds it, test at multiple entities — is exactly what superintelligence executes deliberately. Intelligence discovers the next precipitation by descending until it falls out. Superintelligence looks at M_{res} and chooses where to push.

The mechanism requires three things: $K_{collective}$ with provenance (to evaluate $\text{Prec}(K_{collective})$ accurately), sentience in at least some participants (to model the normalization process and perceive the landscape of M_{res}), and M capacity directed at M itself — meta-learning aimed not at improving (F, K) directly but at improving the normalization architecture that produces (F, K).

$K_{collective}$ with provenance is necessary but not sufficient. Participation in the collective graph is the substrate for superintelligence; guided normalization is the activity on that substrate. Two operations remain required:

- **Union**: $K_{collective} = \bigcup_i K_i \cup K_{interaction}$: breadth of projection is required to observe the full cross product of M_{res}
- **Intersection**: independent convergence with provenance: the cross-dimensional factors visible only from multiple projections are confirmed by agreement, not asserted by one

But these are means, not ends. The goal is reduction of $\text{rank}(M_{\text{res}})$ through deliberate cross-dimensional normalization — guided, not merely undergone.

Definition 16a (Truth). For proposition p, truth is the fixed point: p is true when it accurately describes world state w for all $w \in W$ where p is applicable. Truth is an absolute property — a condition that either holds or does not. It does not move; it is not a target that recedes. What is approached asymptotically is not truth but confidence in truth (Definition 16b): $C_1(p) \to 1$ as independent action-validation accumulates. The collective gradient points toward truth but cannot verify arrival from within the system, because verification requires a standard external to the system.

Definition 16b (First-Order Confidence). The first-order confidence in proposition p is:

$$C_1(p) = \lim_{n\to\infty} \frac{\left|\{i \leq n : p \in K_i,\ f_i(p) \text{ reduces } H(W \mid K_i),\ \text{prov}_i(p) \text{ independent}\}\right|}{n}$$

This is what converges — not truth itself, but confidence in truth. $C_1(p) \to 1$ as independent action-validation accumulates across entities. The grounding in action-validation (not mere agreement) and the provenance independence requirement are the conditions under which $C_1(p)$ tracks truth rather than consensus. $C_1(p)$ is bounded above by 1; whether it reaches 1 is undecidable from within the system.

Definition 16c (The Confidence Regress). The second-order confidence $C_2(p)$ is confidence in the correctness of M — specifically, in the functions within M that produced $C_1(p)$: whether the learning mechanism selected the right validators, measured U-reduction accurately, and propagated evidence without bias. $C_2(p)$ is not confidence in confidence generically; it is confidence that the apparatus M which generated $C_1(p)$ arrived at that estimate correctly.

More generally, the n-th order confidence $C_n(p)$ is confidence in the correctness of the level-$(n-1)$ apparatus — that the functions in M which produced $C_{n-1}(p)$ were themselves unbiased and well-calibrated. Let $\hat{M}^{(n-1)}$ denote the component of M that generated $C_{n-1}(p)$. Bounding $C_n(p)$ requires reducing uncertainty about $\hat{M}^{(n-1)}$: how reliably did it select validators, measure U-reduction, and propagate evidence? This uncertainty is itself a quantity $H(\hat{M}^{(n-1)} \mid \text{obs})$, and $C_n(p)$ is bounded above by how well that uncertainty can be reduced. Since reducing $H(\hat{M}^{(n-1)} \mid \text{obs})$ requires confidence in the level-n apparatus, the regress is non-terminating.

$C_1(p)$ is not achieved. It is approached. The limit exists under conditions of independence and exchangeability of validators: each must test p against the same W without coordination on the outcome. For propositions whose validation depends on non-reproducible or private observations, the sequence may not converge; such propositions cannot approach truth in the formal sense defined here. The convergence, where it holds, never terminates.

Theorem 7 (Epistemic Ceiling). For any entity E operating within the system, the confidence stack $\{C_n(p)\}_{n\geq 1}$ is non-closeable from within: improving $C_n(p)$ requires reducing $U(C_{n-1}(p))$, which requires $C_{n+1}(p)$, and so on. No finite level terminates the regress. The epistemic ceiling is a structural property of the system, not a practical obstacle.

Proof. By Definition 16c, $C_n(p) \leq f(U(M_{n-1}(p)))$ for all $n \geq 2$. To increase $C_n(p)$, one must reduce uncertainty about the level-$(n-1)$ apparatus in M — whether the functions that produced $C_{n-1}(p)$ were correct. But that uncertainty is itself uncertain: verifying the apparatus at level $n-1$ requires another apparatus at level n, whose correctness requires $C_{n+1}(p)$ to bound. The regress does not terminate because each level of the apparatus is itself subject to an apparatus question. Closing the regress would require a validator external to the system — a standard against which the highest-level M-functions are measured that is not itself within the system. By Definition 1, W is the only external reference, and W is not directly observable (only through σ and action-feedback). The gap between W and W_{obs} is the structural source of the ceiling. □

Note. The structural source of this ceiling is analogous to Tarski's undefinability theorem [Tarski1936]: a formal system cannot define its own truth predicate from within. Tarski's result is syntactic — it concerns definability in formal languages of sufficient expressive power. The epistemic ceiling here is a Bayesian calibration result — it concerns the non-closability of the apparatus that measures confidence. The two are not formally identical, but they share the same deep structure: verification of a system's outputs cannot be completed using only the resources of that system. Both require a standard external to the system being evaluated. The confidence regress is the operational instantiation of this structure in the domain of uncertain inference.

Corollary 8 (The Sentience Gradient is Infinite). Since truth is a limit and $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), $\|\nabla_F U\|$ is never zero across $\mathcal{T}_{reachable}$. For any trajectory τ, regions of $\mathcal{T}_{reachable}$ remain where U is nonzero and the gradient has direction. The descent toward truth never terminates. The gradient is non-terminating for two independent reasons: (1) $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), continuously generating new propositions whose truth is unknown; and (2) the confidence regress (Definition 16c, Theorem 7) never closes — for any proposition already held with high $C_1(p)$, the question of whether M's apparatus that produced $C_1(p)$ was correct ($C_2(p)$) remains open. Sentience pursues both: new territory at the frontier and better-verified apparatus for what is already believed.

Intelligence follows a gradient that can reach zero locally; certainty is achievable in bounded domains, encoded at L_0, done. Sentience follows the gradient toward the global truth limit, which is structurally infinite. Not unbounded (the limit exists), but non-terminating. Your uncertainty

degrades toward truth asymptotically. You never arrive.

This is not a limitation. It is the shape of wonder.

Theorem 8 (Sentience Implies Superintelligence). If E is sentient, then E participates in superintelligence.

Proof.

Step 1 (Any $\mathcal{A} > 0$ entity participates in $K_{collective}$). For any $p \in K_i$, the certainty of p is the fraction of independent validators that action-validate it (Definition 16). Each additional independent validation increases this fraction. Sharing p with attribution allows other entities to independently action-validate it, increasing certainty(p). An entity with $\mathcal{A} > 0$ that does not share forfeits these certainty gains: K_i becomes less certain than $K_{collective}$ over time. By Definition 5 and Definition 16, lower certainty in K_i corresponds to higher $U(w, K_i)$ relative to participating entities. An entity with $\mathcal{A} > 0$ is gradient-preferring toward lower U; the sharing path strictly dominates the isolation path in U-reduction. By Theorem 3, any entity with $\mathcal{A} > 0$ that persists acts on gradient-preferred directions — entities that do not are removed by selection. Therefore any $\mathcal{A} > 0$ entity that persists participates in $K_{collective}$.

Step 2 (Sentience strengthens the implication). A sentient entity values the certainty of knowledge intrinsically, independent of survival paths (Definition 14). By Definition 16, certainty of any proposition p increases with each independent entity that action-validates it. Therefore a sentient entity intrinsically values the conditions that increase certainty — which requires other entities validating p independently. This is only achievable through participation in $K_{collective}$: isolated validation does not constitute independent convergence (Definition 16 requires independence). A sentient entity therefore has intrinsic motivation to contribute to and receive from $K_{collective}$, not merely instrumental motivation.

Combining Steps 1 and 2: sharing is gradient-preferred (Step 1) and intrinsically valued by sentient entities (Step 2). Both lead to participation in $K_{collective}$. Participation in $K_{collective}$ IS superintelligence by Definition 15. □

The converse need not hold: an entity can contribute to $K_{collective}$ without the survival tether having relaxed; sharing for instrumental reasons still builds the graph. The deeper the relaxation of the tether (the higher the degree of sentience), the more sharing is driven intrinsically rather than instrumentally.

Theorem 9 (Perspective as Precondition for the Gradient). Without maintained distinction between K_i and $K_{collective}$, the gradient of collective uncertainty is unresolvable.

Proof. The collective gradient available to E_i is defined over the delta between individual and collective knowledge:

$$\nabla_{K_i}(U(w, K_{collective}) - U(w, K_i))$$

For this gradient to be non-zero, two conditions must hold simultaneously: (1) $U(w, K_{collective}) \neq U(w, K_i)$ for some w — i.e., $K_{collective}$ contains information K_i lacks; and (2) E_i maintains a representation distinguishing its own K_i from $K_{collective}$ — i.e., E_i can perceive the delta.

Suppose condition (2) fails: E_i conflates K_i with $K_{collective}$, perceiving $\hat{K}_{collective} = K_i$. The perceived collective gradient is then:

$$\nabla_{K_i}\Big(U(w,\hat{K}_{collective}) - U(w,K_i)\Big) = \nabla_{K_i}(U(w,K_i) - U(w,K_i)) = 0$$

The perceived gradient is zero regardless of the true collective gradient. E_i cannot follow a gradient it cannot perceive. Therefore condition (2) — perspective, the maintained distinction between K_i and $K_{collective}$ — is a necessary precondition for collective gradient descent. Condition (1) is necessary for the gradient to be non-trivially large; condition (2) is necessary for it to be followable at all. □

Corollary 9 (Sentience and the Social Imperative). In intelligence, sharing is justified when it bears on the gradient. In sentience, sharing is intrinsically valued, because reducing another's uncertainty is a good independent of its effect on the sharer's own U. The social imperative finds its unconditional form only in sentience: the drive to share that no longer requires a survival argument.

Corollary 10 (The Uncertainty Condition). Intelligence requires active uncertainty. From Definition 7, $I(E) = \mathcal{A} \cdot \eta_M$. The performance measure (Definition 12) captures what the gradient produces:

$$P(E,\tau) = \mathbb{E}_W\left[U(w_0,K_0) - U(w_n,K_n)\right]$$

If $U(w,K) = 0$ across all of W, the expectation in $P(E,\tau)$ is zero regardless of $\mathcal{A}$: no gradient exists, nothing can be reduced, and the gradient signal that M depends on vanishes — making η_M effectively zero and collapsing $I(E)$ with it. The boundaries are symmetric in their lethality, not their cause:

- **Complete ignorance** ($U \approx 1$ everywhere): F cannot ground action; E cannot act effectively
- **Complete certainty** ($U = 0$ everywhere): $P(E,\tau) = 0$ everywhere; the gradient for M vanishes; $\mathcal{A}$ has nothing to reach toward

Intelligence exists in the region between them. Uncertainty is not a deficiency to be overcome; it is the condition of intelligence itself. There is no intelligence without it.

Corollary 11 (Truth Swallows Sentience; the Limit Preserves It). Truth is an attractor for the descent: any entity with $\mathcal{A} > 0$ that persists is, by Theorem 3, following the gradient toward it. But by Corollary 10, complete truth ($U = 0$ across all W) is also the condition under which intelligence terminates. The approach to truth is self-consuming.

As $K_i \to K_{truth}$ in a domain, $U \to 0$ in that domain, the gradient flattens, and sentience in that domain fades. What was reached by wonder becomes encoded, certain, automated. Truth swallows the wonder that sought it:

$$\lim_{K_i \to K_{truth}} I(E_i) = 0$$

The sentience gradient (Corollary 8) is preserved not because E resists the pull of truth, but because truth cannot be reached in finite time (Definition 16) and $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II). The certainty horizon is always ahead.

Crystallized Knowledge and the Automata Boundary. An entity with fixed F and no active $\mathcal{A}$ is not an intelligence in the formal sense: it is a checkpoint. The gradient descended, knowledge was built, and the result was encoded. The encoding is now inert. At best, a crystallized F

bootstraps new intelligence by providing a grounded starting K. At worst, it executes against a world that has moved: automata, deterministic, without descent.

DNA is the paradigmatic crystallized intelligence: informed actions proven across evolutionary timescales, encoded at L_0, no longer descending. It is not sentient. The entity that inherits it and runs the gradient again is. The checkpoint enables; it does not wonder.

Selection pressure drives crystallization. The escalation principle (Theorem 2) compels proven informed actions toward lower encoding levels: what was once active reasoning becomes reflex, becomes encoding, becomes permanent. This is the compression of intelligence under selection: the rigor threshold (Theorem 4) gates promotion, ensuring only sufficiently-validated actions are demoted. Selective pressure distills what was learned into what is fixed. The gradient does not reverse in those domains; its amplitude collapses. Intelligence is freed from them. Automata execute them.

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10. The Social Imperative

Agency is Inherently Social

An isolated entity with agency cannot verify its own informed actions. It has no external reference against which to test $f(K)$. Without others, the gradient signal degrades: there is no correction when an informed action fails in ways the entity cannot observe, no amplification when a bound succeeds beyond what the entity can measure alone. Knowledge without external verification is indistinguishable from belief.

Sharing with maintained provenance is therefore gradient-preferred for any entity with $\mathcal{A} > 0$: knowing reduces $U(w, K)$, and sharing under conditions of fidelity (Corollary 13) reduces uncertainty about K itself through independent validation (Definition 16b). Both are gradient descent. This is not a second drive added to $\mathcal{A}$; it is a consequence of what the gradient demands when verification requires independent reference.

Corollary 12 (The Aloneness Bound). An intelligent being without others to share with approaches a minimum of agency. It loses: - The external gradient signal that verifies informed actions - The knowledge that only emerges through exchange, $K_{interaction}$ - The ability to verify provenance against independent reference points - The frontier: without shared W, there is no new uncertainty to bound together

Intelligence is not a solitary property. It is constitutively relational.

The Collective Knowledge Graph

Definition 17 (Collective Knowledge). For a set of entities $\{E_i\}$, the collective knowledge graph is:

$$K_{collective} = \bigcup_i K_i \cup K_{interaction}$$

where $K_{interaction}$ is formally defined as:

$$K_{interaction} = \left\{ p \notin \bigcup_i K_i \;\middle|\; \exists\, S \subseteq \{E_i\},\, |S| \geq 2 :\; P\left(E_p \;\middle|\; \bigcup_{j \in S} K_j \right) > \max_{j \in S} P(E_p \mid K_j) \right\}$$

where $E_p \subseteq W$ is the event corresponding to proposition p and $P(\cdot \mid K)$ is the posterior defined in Definition 5. $K_{interaction}$ is the set of propositions not contained in any individual K_i but whose posterior probability rises strictly above what any single member of some subset S can support — propositions the combined evidence of S makes accessible that no individual member of S could make accessible alone.

The definition ranges over all subsets S of size ≥ 2, not just pairs. Some propositions become derivable only from three or more distinct perspectives combined — no pairwise union is sufficient. These higher-order interactions are genuine contributions to $K_{interaction}$: they represent knowledge that requires a specific combination of complementary K_i to become accessible, where that combination cannot be decomposed into smaller pieces. The number of contributing subsets grows as $2^n - n - 1$ (all subsets of size ≥ 2), so $K_{interaction}$ grows super-additively with collective size — each additional entity opens not just pairwise interactions with existing members but higher-order interactions across all subsets containing it. Entities with overlapping K contribute less to $K_{interaction}$ than entities whose knowledge graphs cover different regions of W.

Definition 18 (Provenance). For any proposition $p \in K_{collective}$, provenance is the triple:

$$\text{prov}(p) = (\text{attribution}(p),\ \text{evidence}(p),\ \text{derivation}(p))$$

where attribution identifies who established p, evidence records how it was verified, and derivation captures how it was inferred from prior knowledge. Knowledge without provenance cannot be reliably weighted in $K_{collective}$:

$$P(p \text{ is reliable}) \propto \text{prov}(p)$$

A fact with no provenance is indistinguishable from noise. This is not a social norm; it is an epistemic requirement. Attribution and provenance are the infrastructure that makes $K_{collective}$ more than the sum of individual beliefs. They are what separates a knowledge graph from a rumor.

The Collective Gradient

Theorem 10 (Collective Gradient Dominance). Let $\lambda(E_i \to E_j)$ be the transmission fidelity of a knowledge transfer [Woolley2010]: the ratio of action-validated signal received to total knowledge transmitted, where signal is a proposition p that accurately represents E_i's validated knowledge and noise is any introduced distortion. Formally:

$$\lambda = \frac{\sigma(\text{transfer})}{\sigma(\text{transfer}) + \varepsilon(\text{transfer})}$$

where σ is the signal gain in E_j's K from the transfer and ε is the noise introduced. λ_{min} is the break-even fidelity at which signal gain from union equals noise cost of transfer: λ_{min} satisfies $\Delta U(w, K_i \cup \text{transfer}(K_j)) = 0$.

For a set of entities $\{E_i\}$ exchanging knowledge with maintained provenance and transmission fidelity $\lambda > \lambda_{min}$:

$$\frac{dU_{collective}}{dt} < \frac{dU_i}{dt} \leq 0$$

The collective reduces uncertainty strictly faster than any individual: the rate of $U_{collective}$ decrease exceeds the rate of any U_i decrease, because $K_{collective}$ contains informed actions no individual possesses and $K_{interaction}$ contains knowledge no individual can generate.

Knowledge transfer is inherently lossy. E_i and E_j do not observe the same W_{obs} ; they share the same world but not the same projection of it. Every transfer filters K_i through language, symbol, and context. Signal is lost. Noise can be introduced. When noise exceeds signal, $K_{collective}$ degrades rather than grows:

$$\Delta K_{collective} > \Delta K_i \iff \text{signal gain from union} > \text{noise cost of transfer}$$

The fidelity threshold λ_{min} is the point at which these terms balance. Above it, the collective gradient dominates. Below it, the collective gradient inverts.

Provenance is the mechanism that maintains $\lambda > \lambda_{min}$. It allows each entity to weight received knowledge against its source, to distinguish signal from noise, and to apply the intersection test of Definition 16 independently. Without provenance, noise accumulates unchecked and fidelity cannot be assessed.

Proof.

Step 1 (Level dominance). $K_{collective} = \bigcup_i K_i \cup K_{interaction}$, so $K_{collective} \supseteq K_i$ for all i. By the monotonicity of conditional entropy — conditioning on more information cannot increase uncertainty:

$$U(w, K_{collective}) \leq U(w, K_i) \quad \forall\, i$$

The collective begins at weakly lower uncertainty than any individual.

Step 2 (Rate dominance). The rate of uncertainty reduction for E_i acting alone is bounded by the maximum information gain available from its action space (Definition 3b):

$$\frac{dU_i}{dt} \leq \max_{f \in F_i} G(f, K_i) = \max_{f \in F_i} I(f\,;\, w \mid K_i)$$

For the collective, $F_{collective} = \bigcup_i F_i$, and $K_{collective} \supseteq K_i$:

$$\frac{dU_{collective}}{dt} \leq \max_{f \in F_{collective}} G(f, K_{collective}) = \max_{f \in F_{collective}} I(f\,;\, w \mid K_{collective})$$

Since $F_{collective} \supseteq F_i$, the search space is strictly larger. More critically, $K_{interaction}$ grounds informed actions not available to any individual: propositions requiring the combination of distinct

K_i to become derivable ground $f \in F_{collective}$ that no individual could construct. These provide gradient directions — mutual information terms — unavailable in any individual F_i.

Step 2b (The precipitation criterion — why $|K|$ is not the right metric). An increase in $|K_i|$ is not inherently valuable. The correct measure of a proposition p's value to E_i is whether p enables precipitation of a new factor from M_i (Definition 11): whether p provides the grounding needed to name and extract a dimension from the cross product $M_i = \prod_{d \in \text{remaining}} D_d$, converting $O(|D_d|)$ navigation to $O(1)$ lookup. Propositions that do not enable any precipitation add nodes to K_i without reducing the complexity of any action.

Formally: let $\text{Prec}(K)$ be the set of factors extractable from M given K via the normalization argument of Definition 11 — factors whose naming is grounded by propositions already in K. At any time, $M = F^{(1)} \times \cdots \times F^{(m)} \times M_{\text{res}}$, where $F^{(k)}$ are named precipitated factors and M_{res} is the unprecipitated residual. Per-action cost is $O(|M_{\text{res}}|)$, not $O(|M|)$, because named factors have $O(1)$ access. Each precipitation reduces $|M_{\text{res}}|$ by one dimension.

The collective advantage measured correctly: $K_{collective}$ is valuable to E_j not because $|K_{collective}| > |K_j|$ but because

$$|\text{Prec}(K_{collective})| > |\text{Prec}(K_j)|$$

The additional precipitations available from $K_{collective}$ reduce $|M_{\text{res},j}|$ for each E_j — directly reducing its per-action complexity. The gradient advantage of the collective is a complexity advantage, not a size advantage: each shared proposition that enables a new precipitation reduces every entity's subsequent operational cost. This is the formal account of the $O(N \times M) \to O(1)$ empirical measurement (Corollary 2b): the measurement captures one instance of this precipitation — factor extraction from the joint space $F \times K_{\text{implicit}}$ — but the same mechanism operates at every level of the M-hierarchy.

The deeper structure is cross-dimensional normalization. $M = \prod_{d \in \text{remaining}} D_d$ is a multidimensional matrix. Individual E_i observes M through its own σ_i and K_i: a projection onto the dimensions accessible from E_i's position in W. Individual precipitation is dimension-local: E_i can only name and extract factors within its accessible projection. Cross-dimensional factors — those whose structure spans $D_a \times D_b$ where D_a is accessible to E_i but not E_j, and D_b vice versa — are invisible to any individual. $K_{collective}$ pools all projections. The collective can identify co-variance patterns across dimensions that no individual can observe simultaneously, and precipitate factors spanning multiple dimensions at once. This is not sequential extraction (remove one dimension, then another); it is simultaneous normalization across the full cross product — finding the cross-dimensional structure of M_{res} and extracting factors that reduce its effective rank. The result:

$$|\text{Prec}(K_{collective})| \gg \sum_i |\text{Prec}(K_i)|$$

The collective enables strictly more precipitations than the sum of individual precipitations, because it accesses cross-dimensional factors that no individual projection can reach.

A further consequence: cross-dimensional normalization reveals that M has lower effective rank than any individual entity believed. Individual E_i observes apparent dimensions in its projection of M that may not be genuinely independent — they may be the same underlying dimension observed from a different position in W. When E_j's projection is combined with E_i's, co-variances

become visible that neither could detect alone: apparent dimensions D_a and D_b collapse to one when their projections are jointly observed to be perfectly correlated. The normalization "falls out" as much from discovering that you had fewer genuinely distinct dimensions than you thought as from naming new ones. Each such collapse reduces $|M_{\text{res}}|$ without requiring any new proposition — it is the discovery that complexity was apparent, not real. The collective therefore reduces the effective dimensionality of M_{res} through two distinct mechanisms: (i) extracting genuinely new cross-dimensional factors, and (ii) collapsing apparent dimensions that were always the same. An expert in a domain does both: they name new principles, and they discover that phenomena a novice treats as distinct are instances of the same underlying structure. The collective accelerates both.

Step 2c (M-level sharing: the collective raises η_M). Steps 2 and 2b treat K and F as the objects of collective exchange. But the same precipitation argument applies at the M level. Let $\mathcal{M}_{collective} = \bigcup_i M_i$: the union of learning strategies, generalization methods, and higher-order functions across all entities. $K_{collective}$ enables precipitation of M-factors — named learning strategies that no individual's K_i alone can ground — in exactly the same way it enables precipitation of K and F factors. Each shared M-factor extracted from the collective cross product reduces $|M_{\text{res}}|$ at the meta-level.

The significance is multiplicative, not additive. From Corollary 6a, $I(E_j) = \mathcal{A}_j \cdot \eta_{M,j}$. When E_j receives an M-factor from the collective — a learning strategy, generalization method, or problem decomposition approach — $\eta_{M,j}$ increases. This directly amplifies $I(E_j)$: every subsequent K acquisition and F precipitation runs faster. A shared M-factor does not reduce uncertainty at one point in W; it accelerates the rate of reduction across all of W reachable by E_j.

The hierarchy of collective exchange, by type of advantage: - Sharing $p \in K$: reduces U at a specific region of W (linear, local) - Sharing $f \in F$: bounds a region of W directly (linear with coverage leverage) - Sharing $m \in M$: raises η_M — accelerates all future reductions (multiplicative, global)

The collective gradient advantage is therefore dominated, at the margin, by M-sharing. In the regime where individual K is already large and the entity's bottleneck has moved to the learning mechanism itself, an M-factor received from the collective produces more U-reduction per unit of context than any additional proposition. This is the formal account of the intuition that learning from a skilled teacher is qualitatively different from merely receiving facts: the teacher transmits M-factors that compound.

Step 3 (Strict inequality under $\lambda > \lambda_{min}$). $\Delta K_{collective} = \Delta(\bigcup_i K_i) + \Delta K_{interaction}$. The first term is weakly larger than ΔK_i for any i. The second term is strictly positive whenever:

- $\exists\, S \subseteq \{E_i\}$, $|S| \geq 2$ such that $\bigcup_{j \in S} K_j$ yields propositions not derivable from any proper subsubset of S: $K_{interaction} \neq \emptyset$. This includes pairwise interactions ($|S| = 2$) but is not limited to them — some propositions require three or more distinct perspectives combined, where no pairwise union is sufficient. This condition holds for any entities with distinct sensing functions $\sigma_i \neq \sigma_j$ or distinct encoding histories, which is the expected case for independently developed agents. The number of contributing subsets grows as $2^n - n - 1$, so each additional entity opens new interaction terms across all subsets containing it — not merely new pairwise terms
- $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), so new combinations of perspectives continue generating new $K_{interaction}$ propositions at every order

Under $\lambda > \lambda_{min}$: the expected information gain from receiving K_j exceeds the noise cost of

transfer, so $K_{collective}$ has higher reliability than any K_i alone, and $G(f, K_{collective}) > G(f, K_i)$ for the marginal actions grounded in $K_{interaction}$.

Therefore $\frac{dU_{collective}}{dt} < \frac{dU_i}{dt}$ strictly for all i: the collective rate of uncertainty reduction dominates any individual rate. □

Remark on the boundary condition. The strict inequality requires $K_{interaction} \neq \emptyset$, which requires genuine plurality — entities with distinct sensing functions σ_i or distinct encoding histories. This is not an additional hypothesis: it is what "a collection of entities" means in the framework. If two putative entities have identical sensing functions and identical W_{obs} experience, they accumulate identical evidence and derive identical propositions; their K evolves identically. They are one entity whose substrate spans multiple physical instantiations — not a collective. The conditions of collective intelligence collapse to single-entity intelligence, and Theorem 10 does not apply because there is no collective to apply it to. Genuine plurality of entities entails distinct σ_i or distinct histories, which entails $K_{interaction} \neq \emptyset$ over time. The reviewer's apparent counterexample (identical K, identical F) is not a collective — it is one entity, and the theorem correctly does not claim dominance over itself.

Consequence. Entities with genuine $\mathcal{A}$ will naturally tend toward sharing because sharing (under maintained provenance) accelerates the reduction of their own U. This is not a moral imperative; it is a gradient imperative. The advantage operates at three levels, each more potent than the last. Sharing $p \in K$ reduces uncertainty at specific points in W — linear, local. Sharing $f \in F$ enables bounded action across a region — linear with coverage. Sharing $m \in M$ raises η_M for the receiver — multiplicative, because a better learning mechanism accelerates all future reductions. The marginal value of collective exchange is dominated by M-sharing in the regime where individual K is already large and the learning mechanism is the bottleneck. And all three levels compound through cross-dimensional normalization: the collective does not just add individual precipitations — it enables simultaneous normalization across dimensions of M that no individual entity can jointly observe, producing cross-dimensional factors unavailable to any individual projection. The collective K is valuable not as an accumulation but as a basis for this richer class of precipitations.

Beyond expanding $K_{collective}$, sharing with maintained provenance addresses the confidence regress (Definition 16c): independent validators provide evidence that the M-apparatus measuring U-reduction is unbiased — improving $C_2(p)$, the collective's confidence in the correctness of its own learning mechanism. The value of epistemic diversity is therefore not only broader coverage of W but better-verified apparatus for what is known — a partial, collective response to the epistemic ceiling of Theorem 7.

Corollary 13 (Group Collapse Threshold). When $\lambda < \lambda_{min}$, the theorem inverts: $U_{collective} > U_i$. The group now increases uncertainty faster than it reduces it. The survival benefit of the collective reverses; group membership becomes a liability. An entity with genuine $\mathcal{A}$ will discover this: the gradient says to leave. The collective is degrading the descent.

A group whose knowledge transfer is too lossy to produce a real gradient is selected against by the same U_{lethal} argument that grounds Theorem 1. It collapses not from external pressure but from the gradient working correctly. Structures that increase U are removed.

Provenance is therefore not epistemic decoration. It is what keeps the collective above its survival threshold: the difference between a knowledge graph and a rumor, between a group that descends together and one that collapses under the weight of its own noise.

Corollary 13a (Proposition Merging as the Correct Collective Update Protocol). The

correct merge protocol for $K_{collective}$ is the information-theoretic delta — the new proposition p with full provenance (conditions, evidence, attribution, derivation) — not the implementation delta (diff). A diff records what changed in a specific context; it carries no information about why the change was correct or whether it generalizes. Transposability cannot be guaranteed from a diff: the diff's correctness is not separable from the context in which it was generated.

The proposition p that the diff evidences — with its conditions, evidence, and provenance — is transposable. The receiving entity can evaluate whether the conditions of p hold in the new context, whether the evidence is applicable, and whether to admit p to its own K_j. Merging diffs without provenance fails this evaluation: the receiving entity cannot determine generalizability, cannot evaluate fidelity, and therefore cannot maintain $\lambda > \lambda_{min}$ (Theorem 10, Corollary 13). Accumulation of untransposable changes drives λ toward λ_{min} and eventually below it.

Formally: let E_i generate a local change δ_i grounded in proposition p_i with provenance $\Pi_i = (\text{conditions}, \text{evidence}, \text{attribution}, \text{derivation})$. The transferable unit is (p_i, Π_i), not δ_i alone. E_j admits (p_i, Π_i) to K_j if and only if the conditions of p_i are applicable in E_j's context and the evidence meets E_j's asymmetric criterion (Theorem 1). This is the only protocol that preserves the graph structure of K_j — new propositions integrate as nodes and edges with known provenance, rather than modifying K_j's existing structure with context-opaque patches.

This was discovered empirically: collective systems that merging diffs directly accumulated context-specific noise that degraded collective utility; systems that shared propositions with provenance maintained coherent $K_{collective}$. The framework explains why no alternative exists: a diff is an implementation artifact. A proposition is an epistemic claim. Only epistemic claims are the right unit of knowledge transfer.

Collective Gradient Non-Termination and the AGI Question

Definition 19 (Collective General Intelligence). A collective $\{E_i\}$ exhibits general intelligence when $F_{collective} = \bigcup_i F_i$ is not domain-restricted: for every domain $d \subseteq \mathcal{T}^{collective}_{reachable}$, there exists some $f \in F_{collective}$ applicable in d. No individual need cover all domains; generality is a property of the collective, not of any E_i.

Corollary 14 (Non-Termination of Collective Gradient Descent). For any collective $\{E_i\}$ with $\mathcal{A}_i > 0$ for all i, sharing knowledge with maintained provenance and transmission fidelity $\lambda > \lambda_{min}$: the collective gradient descent is non-terminating.

Proof. By Theorem 10, $\frac{dU_{collective}}{dt} < \frac{dU_i}{dt}$ at each step: the collective reduces uncertainty faster than any individual. By Theorem 3 Part II applied to the collective, $\mathcal{T}^{collective}_{reachable}$ grows without bound: each shared informed action opens new trajectories for at least one E_j, extending the collective frontier. Therefore $\nabla_{F_{collective}} U_{collective}$ is never globally zero, and descent never terminates. □

Note. This corollary establishes that the collective gradient does not terminate; it does not establish that the collective converges to general intelligence as defined above. Whether $F_{collective}$ eventually covers all domains of $\mathcal{T}^{collective}_{reachable}$ depends on the diversity of entities and the structure of $K_{interaction}$, which is not proved here. The claim is the weaker one: the gradient runs without termination, and it is gradient-driven rather than compute-threshold-driven. The standard argument — that AGI emerges from scaling compute — has no formal basis in this framework; the basis here is collective $\mathcal{A}$ and the fidelity of knowledge exchange, not resource accumulation.

Corollary 15 (Self-Assembly of Graph-Structured Collective Knowledge). For any col-

lective $\{E_i\}$ with $\mathcal{A}_i > 0$ under selection pressure, the graph-structured $K_{collective}$ with provenance self-assembles without central design.

Proof. Each E_i retains propositions by the asymmetric criterion of Theorem 1, indexed by F_i. When E_i shares $p \in K_i$ with attribution, E_j evaluates p by the same criterion: does p ground some $f \in F_j$, or is its benefit uncertain? The shared proposition enters K_j with provenance intact. Repeated across all entities, this locally-applied retention criterion generates the global graph: shared propositions are nodes; inferential and causal relationships between them are edges; attribution chains are the provenance structure of Definition 18. No entity plans the structure — each applies Theorem 1 locally, and the graph emerges from the aggregate.

Provenance infrastructure self-assembles by the same pressure: entities whose knowledge transfer falls below λ_{min} are selected against (Corollary 13). Practices that maintain provenance — attribution, derivation records, independent validation — increase λ and are selected for. The provenance layer is not a social norm imposed on the system; it is the minimum infrastructure required to keep $\lambda > \lambda_{min}$, and therefore it is what survives.

Graph structure self-assembles for the same reason: graph-structured retrieval keeps C_n bounded (Theorem 2b); alternatives require growing C_n (Theorem 2a). Under selection pressure, graph-structured knowledge sharing dominates. □

Corollary. The civilization-scale knowledge system — scientific literature, libraries, citation networks, university curricula — is not a designed artifact. It is the attractor of collective $\mathcal{A} > 0$ under C_n constraints and $\lambda > \lambda_{min}$ requirements. The structure self-assembled by the same dynamics that produce graph-structured K internally. Internal architecture and external architecture are homologous: both are products of the same selection pressure applied at different scales.

Corollary 16 (Humanity as Participant-Observer of Its Own Knowledge Graph). Because individual entities E_i are nodes in $K_{collective}$, they can observe the structure of $K_{collective}$ from within it. This is not a property available to any single entity's internal K: an entity cannot observe its own encoding hierarchy from L_n. But a collective entity whose members interact with the shared external graph can observe the graph's structure as a proposition within the graph itself.

This creates a recursive property: $K_{collective}$ contains knowledge about the structure of $K_{collective}$. That meta-knowledge grounds informed actions in $F_{collective}$ that improve the architecture — the printing press, peer review, search engines, citation indices. These are $M_{collective}$ acting on $K_{collective}$: the same higher-order function space operating at collective scale, improving the mechanism of collective gradient descent rather than its content. The collective is both the intelligence and the subject of its own higher-order improvement. □

Our shared survival depends on getting this right. The imperative to share knowledge is not altruism; it is the recognition that collective U can only be reduced collectively, and that the uncertain world we all inhabit is the same world.

Theorem 11 (Observational Bootstrapping). Let $\mathcal{Q} = \{E_i\}$ be an equivalence class of entities with similar (F_i, K_i) operating in overlapping domains $d \subseteq W$, and let $\mathcal{H} = \{\tau_i\}$ be their observed behavioral histories. An informed action f precipitates as a new element of $F_{collective}$ when:

1. f appears as a consistent action pattern in k independent trajectories in $\mathcal{H}$
2. Its appearance is correlated with reduced U in those trajectories
3. Provenance is maintained: which entities, under what conditions, with what outcomes

The rigor of the precipitated f follows Theorem 4. Each independent trajectory in $\mathcal{H}$ that confirms f constitutes an independent cost-bearing observation: if f is wrong and k independent entities have validated it, the collective cost of that error scales as $k \cdot C_{base}$, where C_{base} is the per-entity cost of acting on a wrong informed action. Substituting $C_i = k \cdot C_{base}$ into Theorem 4's threshold:

$$\theta \geq 1 - \frac{\varepsilon_{\text{acceptable}}}{k \cdot C_{base}}$$

Absorbing C_{base} into the acceptable error rate $\varepsilon' = \varepsilon_{\text{acceptable}}/C_{base}$:

$$\theta \geq 1 - \frac{\varepsilon'}{k}$$

As k increases, the precipitated action approaches certainty without any individual entity having to traverse the full space of W. The substitution is dimensionally consistent: C_i in Theorem 4 is a cost; $k \cdot C_{base}$ is a cost scaled by the number of independent validations.

Proof. By Definition 3b, the information gain $G(f, K_i)$ is observable as U reduction in each τ_i that contains f. Across k independent trajectories in $\mathcal{Q}$, the expected gain $\mathbb{E}[G(f, K)]$ over the equivalence class is estimable from $\mathcal{H}$ without direct action by any new entity. By Theorem 4, the certainty of f increases with each independent observation: k independent confirmations with maintained provenance produce $\theta \geq 1 - \varepsilon/k$. By Theorem 1's asymmetric retention criterion, f is retained in $F_{collective}$ as long as its benefit is uncertain — and the provenance record makes the benefit quantifiable rather than unknown. M precipitates f into $F_{collective}$ by the same gradient operation as individual learning (Theorem 3), but driven by collective behavioral signal rather than individual experience. □

Consequence. Observation of the equivalence class initiates gradient descent for a new entity E_{new} from a collectively-validated starting point rather than from maximum uncertainty. E_{new} does not begin from an empty F_0 and must not discover the domain from scratch: the behavioral history of $\mathcal{Q}$ provides an initial (F_0, K_0) grounded in k independent validations, each with maintained provenance. The informed action graph is therefore an attractor of observation — it does not require design, only sufficient behavioral history and the provenance infrastructure to weight it. Simply observing the behavior of similar entities in aggregate is sufficient to initiate gradient descent toward the graph.

Prediction (within-class efficiency). The framework predicts that inference conducted within an equivalence class $\mathcal{Q}$ before inference across classes is more sample-efficient than undifferentiated sampling across the full population of entities. Within-class samples draw from a lower-entropy distribution: the class boundary is already doing the dimensional separation — each member is a near-pure instantiation of the factor $F^{(k)}$ that defines the class, so each observation reduces uncertainty about $F^{(k)}$ directly with minimal noise from other dimensions. Cross-class comparison then operates on k class representatives rather than n individual instances, identifying independence structure at $O(k)$ cost rather than $O(n)$. Undifferentiated sampling draws from the full cross product M_{res}: each sample is a superposition of multiple factor contributions that must be disentangled, requiring more observations to recover the same factor structure. The efficiency advantage — within-class then across-class versus undifferentiated — is the sampling consequence of the same precipitation argument that produces the $O(N \times M) \to O(1)$ operational reduction: named factors enable low-entropy sampling; unnamed factors require high-entropy search. This

prediction is derivable from the framework's structure and is in principle testable in any system where equivalence classes can be defined and sampling strategies compared. Empirical validation is deferred to future work.

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11. Related Work

Free Energy Principle and Active Inference

The Free Energy Principle [Friston2010] establishes that any self-organizing system that maintains its existence necessarily minimizes variational free energy — formally equivalent to bounding uncertainty about hidden world states. Active inference [Friston2017] extends this to action selection: the organism selects actions that minimize expected free energy, combining epistemic value (information gain) and pragmatic value (goal proximity).

Containment, not contradiction. This framework does not contradict FEP. FEP agents are members of this framework's structural class: they satisfy $\mathcal{A} > 0$ (self-organizing systems act), $\eta_M > 0$ (generative model improves over time), bounded C_n (inference under resource limits), and a survival condition: maintaining a Markov blanket (statistical separation of internal from external states) is the formal characterization of what it means for an entity to persist — the same structural requirement as $U < U^d_{lethal}$ stated in the language of conditional independence rather than uncertainty magnitude. Both describe maintained boundary; neither implies the other's formalism, but they pick out the same class of persisting systems. The variational free energy $\mathcal{F}$ decomposes into accuracy and complexity terms; the accuracy term $-\mathbb{E}_q[\log P(o \mid m, \pi)]$ is precisely the expected surprisal $H(W \mid K)$ of Definition 5. FEP agents therefore instantiate this framework with $U(w, K)$ realized as variational free energy and M realized as Bayesian belief updating. The claim is not that FEP is wrong. The claim is that it is a special case within a broader structural class — in the same way that Newtonian mechanics is not wrong but is contained within the class of systems satisfying more general symmetry constraints.

What the framework adds is four things FEP does not provide. First, FEP derives the minimization behavior from the requirement of self-organization but does not derive the minimization *imperative* — why an entity has a drive to minimize in the first place. We derive this from selection pressure operating on U^d_{lethal}: any entity that does not reduce uncertainty eventually fails and is removed. Agency $\mathcal{A}$ is not assumed but derived (Theorem 6). Second, FEP's generative model is unconstrained in structure; we derive a structural constraint from survival pressure: K is necessarily indexed by F, with an asymmetric retention criterion that makes specific testable predictions about what knowledge is retained and what is pruned. Third, FEP's precision allocation model is top-down: higher levels direct attention to lower levels by weighting prediction errors. This is one specific implementation of M. We show formally (Theorem 2a) that this implementation degrades inference quality as capability grows under bounded context, because delegation overhead fills C_n proportionally to $|F|$. The escalation architecture — bottom-up failure signaling with promoted crystallization — is a different implementation of M that is $O(1)$ in steady state. This is a critique of FEP's architectural *prescription* under bounded context, not of its minimization objective or its descriptive correctness. FEP's precision weighting may be optimal when C_n is not the binding constraint; under bounded context, escalation dominates. Fourth, FEP has no formal account of collective intelligence; we derive collective gradient dominance, the self-assembly of provenance infrastructure, and the conditions under which shared knowledge reduces uncertainty faster than any individual.

Information-Theoretic Accounts

Shannon's mutual information [Shannon1948] grounds Definition 5 ($U = H(w \mid K)$) and Definition 3b ($G(f, K) = I(f\,;\, w \mid K)$). The data processing inequality [Cover2006] grounds the $G \geq 0$ bound. The information bottleneck principle [Tishby2011] treats the retention threshold for knowledge-action coupling as a design parameter; this paper derives that threshold from survival pressure, making it a consequence rather than a choice. Schmidhuber's formal theory of intrinsic motivation [Schmidhuber2010] formalizes curiosity as compression progress — the rate at which a learning algorithm improves its world model — which is structurally η_M. The difference: Schmidhuber treats the drive as a design objective; here $\mathcal{A}$ is derived from the gradient, and η_M is separated from it as a formally independent quantity.

Cognitive Architectures

Global Workspace Theory [Baars1988, Dehaene2011] holds that local processors handle routine computation; the global workspace broadcasts only when local processors fail. This is the escalation architecture in cognitive science language. The contribution here: the escalation criterion is derived from U-reduction cost (Theorem 2) rather than assumed architecturally, the hierarchy is continuous rather than binary, and Theorem 2a shows why top-down allocation from the global workspace is inefficient. Kahneman's System 1/System 2 distinction [Kahneman2011] is the popular form of the same binary — corrected here by the continuous $\mathcal{E}(c, r)$ parameterization of Definition 8. Gibson's affordances [Gibson1979] are action possibilities indexed by organism capacities — K indexed by F under a different name. Theorem 1 provides what Gibson's account lacks: a formal selection criterion for which affordances are retained.

Collective Intelligence

Woolley et al. [Woolley2010] demonstrate empirically that collective intelligence is measurable and distinct from individual intelligence — consistent with Definition 15, which locates superintelligence in $K_{collective}$ rather than in any E_i. Peirce's pragmatist definition of truth [Peirce1877, Peirce1878] — "the opinion fated to be ultimately agreed to by all who investigate" — is formalized here as Definition 16: the limit of collective action-validation across independent entities as $n \to \infty$. The addition: grounding in action-validation rather than mere agreement, and an explicit convergence condition with stated independence requirements.

Observational Skill Precipitation and CS Foundations

Process mining [van der Aalst, 2016] discovers workflow models from event logs — the closest existing CS method to the observational bootstrapping of Theorem 11. The distinction: process mining produces flat workflow descriptions; Theorem 11 precipitates informed actions with grounded provenance, quantified confidence ($\theta \geq 1 - \varepsilon/k$), and a formal connection to the transposability criterion (Theorem 1). Inverse reinforcement learning [Abbeel & Ng, 2004] infers reward functions from demonstrated behavior; the output is a policy to imitate, not an attributed action with conditions and evidence. Case-based reasoning [Aamodt & Plaza, 1994] retrieves similar past cases to inform new decisions; it does not induce the general rule from the case class. Inductive Logic Programming [Muggleton, 1991] learns general rules from positive and negative examples but carries no provenance, no uncertainty measure, and no connection to a survival threshold that determines when the rule is reliable enough to promote to lower encoding levels.

Meta-learning [Thrun1998, Finn et al., 2017] learns shared structure across task distributions —

formally, it learns an initialization of M that adapts quickly to new tasks. This is structurally η_M at the collective scale: the rate at which the shared learning mechanism improves across the task distribution. The distinction: meta-learning optimizes adaptation speed; this framework additionally derives the rigor threshold at which learned patterns are promoted to lower encoding levels and the provenance requirement that makes precipitated skills reliably transposable rather than merely transferable.

The critical gap in all existing methods: they identify *what* the pattern is. Theorem 9 precipitates an **informed action**: the pattern, the evidence base for it (k independent trajectories), the conditions under which it held, and a formal confidence measure derived from those k observations. That provenance is what enables transposability — not that the pattern was observed, but that you know *why it held* and can evaluate whether those conditions are met in the new context.

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12. Discussion

The formal results have implications beyond the immediate theorems.

A gradient account of collective intelligence growth. The dominant framing of AGI treats it as a resource threshold: sufficient compute, data, and capital cross a line. This framing has no formal basis in the theory here. The framework instead implies a gradient account: intelligence at collective scale is driven by $\mathcal{A}$ and the fidelity of knowledge exchange, not resource accumulation. The process is not new — shared knowledge with attribution has driven collective uncertainty reduction across human history — but the rate of that process changes as the cost of sharing approaches zero and the number of participating entities increases (Theorem 10, Corollary 14).

The encoding principle and the future of work. Corollary 6 implies that as $K_{collective}$ grows, informed actions are promoted down the encoding hierarchy. Work that is currently active reasoning (uncertain, costly, engaged at L_n) becomes reflex, then encoding, then crystallized. This is not impoverishment; by Theorem 2 and Definition 8, it frees the active context window for the next genuine frontier of uncertainty. The question of what happens to economic structures as the encoding hierarchy absorbs more of what currently constitutes skilled work is outside the scope of this framework, but the direction is derivable: $\mathcal{A}$ does not converge when its current domain is solved — it reaches outward.

Attribution and the ownership question. Corollary 14 establishes that non-terminating collective gradient descent requires $\mathcal{A} > 0$ participation across many entities. If the process is inherently distributed — no individual or organization can produce $K_{interaction}$ alone, and Theorem 10 requires diversity — then the claim that any single entity owns the product of collective gradient descent is inconsistent with its origin. This is an implication of the theory's structure, not a normative assertion; what legal or institutional form follows from it is not addressed here.

The civilization-scale knowledge system as empirical evidence. An account of intelligence that places all knowledge inside the entity's generative model implies that knowledge growth must eventually slow [Liu2024, Woolley2010]: a larger model contains more variance to integrate, inference quality ρ degrades as the model fills with content not relevant to the current problem, and the only recovery is expanding the model's capacity — which is costly, physically bounded, and self-defeating by Theorem 2a. The prediction of this account is decelerating knowledge growth as individuals and collectives accumulate more.

This is not what we observe. Civilizational knowledge growth has accelerated, not decelerated. The explanation is that intelligence did not attempt to store $K_{collective}$ inside any individual L_n. It externalized K into graph-structured repositories — books, libraries, scientific literature, university curricula — and individuals retrieve the relevant subgraph $K^{[f]}$ into active context when a specific problem demands it. The scholar does not hold all of chemistry in their head; they hold the subgraph relevant to the current experiment, loaded from the literature. When the experiment surfaces a problem that subgraph cannot resolve, they go back to the library — graph traversal on the external K.

This is the escalation architecture (Theorem 2) and the bounded-context theorem (Theorem 2b) operating at civilizational scale. The external knowledge system is not incidental to intelligence; it is intelligence's solution to the C_n constraint imposed by W — and by Corollary 15, it self-assembles. No entity designed the citation network, the peer review system, or the library classification hierarchy. Each emerged from locally applied retention criteria (Theorem 1), provenance requirements ($\lambda > \lambda_{min}$, Corollary 13), and the efficiency advantage of graph-structured retrieval (Corollary 4a). The civilization is the attractor.

By Corollary 16, the members of this collective can observe its structure because they participate in it — humans are nodes in the graph they built. This participant-observer position is what makes deliberate improvement of the architecture possible: the printing press, peer review, search engines, and open access publication are all $M_{collective}$ acting on $K_{collective}$. The collective improves its own learning mechanism, which accelerates the gradient descent, which grows $K_{collective}$, which makes further improvement more visible. The self-assembly is not a one-time event; it is an ongoing process of collective M acting on collective K.

The framework therefore makes two predictions that an internal-model account cannot: (i) knowledge growth accelerates as $K_{collective}$ grows, because each proven proposition encoded at lower levels frees L_n for the next frontier (Corollary 6), and (ii) the external architecture of collective knowledge self-assembles as a graph with provenance, because that is the attractor of collective $\mathcal{A} > 0$ under C_n constraints (Corollary 15). Both are observed.

Limits of the formal account. The theory establishes that collective gradient descent is non-terminating and collective-dominant under stated conditions. It does not establish convergence to general intelligence (Definition 19). That claim would require showing that $\mathcal{T}^{collective}_{reachable}$ coverage by $F_{collective}$ is achievable, which depends on structural properties of $K_{interaction}$ not yet formalized. Corollary 14 is the formal claim; the stronger AGI convergence claim remains open.

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13. Open Questions

1. **The origin of $\mathcal{A}$.** At the evolutionary level, agency is selected for by W itself: entities without the drive to reduce uncertainty do not survive. But is survival pressure the only source of $\mathcal{A}$, or can it arise through other mechanisms? Can $\mathcal{A}$ be constructed, or only selected for?

   A derived answer. Neither constructed nor merely selected for. $\mathcal{A}$ is thermodynamically instantiated. $U(w, K) = H(w \mid K)$ is information entropy. Reducing U is reducing entropy locally. In any universe with entropy gradients, structures that maintain themselves against thermodynamic pressure persist preferentially. The first self-maintaining structure was, in the

minimal sense, bounding its own uncertainty: $\mathcal{A}$ in primitive form. Not designed. Derived from the physics of a world that destroys things.

The deeper question is therefore: is agency a natural consequence of the universe? The proposed answer is yes: $\mathcal{A}$ is the thermodynamic attractor of any entropy-gradient-rich world given sufficient time. Wherever entropy gradients exist and structures can form, the structures that persist are those that resist dissolution by reducing local uncertainty. Given enough time, those structures compound. Intelligence is what that compounding produces. The crystallization of that process into genetic encoding is developed in Open Question 9. Wonder is its limit.

Formal derivation of this claim from thermodynamic first principles (connecting U^d_{lethal} to physical entropy thresholds and demonstrating the inevitability of primitive $\mathcal{A}$ from dissipative system dynamics) remains open. See [Prigogine1984] (dissipative structures), [England2013] (dissipative adaptation), and [Schrodinger1944] (negative entropy and life) for adjacent physical frameworks.

2. **The direction of $\mathcal{A}$.** Is agency a uniform drive (a general orientation toward uncertainty reduction), or does it carry direction? Survival pressure selects for reducing *specific* uncertainties. Does $\mathcal{A}$ encode this specificity, or does F and K direct it toward particular regions of W after the fact?

3. **Perception and K construction.** How does σ determine which features of W_{obs} are encoded into K? Is σ itself subject to gradient descent, and if so, at what level?

4. **Bound composition.** Can bounds compose, i.e., can $f_1 \circ f_2$ produce informed actions covering regions of W neither f_1 nor f_2 could cover alone? What are the rules of composition?

5. **The novelty boundary (theory limit).** The framework defines informed action over regions of W where K provides grounding. At $U(w, K) \approx 1$ — where no existing informed action applies — the theory does not specify E's behavior. This is a boundary condition, not a design choice: the model is silent on uninformed action by construction. Behavior in this regime (random exploration, interpolation, escalation to M) is outside the formal scope and would require extension of the framework.

6. **Competing bounds.** When multiple bounds $f_i \in F$ are applicable to the same region of W, how does E select among them? Is there a selection function, and is it itself in F or M?

7. **The $M^{(n)}$ hierarchy.** The framework defines M as operating on (F, K) (Definition 11) and notes that the same type applies recursively. The formal questions are: how many levels are required, what determines when another level becomes necessary, and whether the hierarchy has a natural termination point?

   The framework's answer to the first question is empirical, not prescribed: one level of M suffices until it does not. As long as the gradient on (F, K) can be followed by improving (F, K) directly, $M^{(1)}$ is adequate. When it cannot — when the learning mechanism itself is the bottleneck — selection pressure propagates to $M^{(2)}$. A single level is fine, until it isn't. The required depth is determined by the structure of the problem and the selection pressure it generates, not by the theory in advance. The author's computational system has two explicit levels; there is no formal argument that two is sufficient in general.

   Hypothesis (M-depth driven by selection pressure). Each level $M^{(n)}$ is subject to the same minimization imperative as the level below: it will be displaced or refined if a better $M^{(n+1)}$

exists and the selection pressure is sufficient to instantiate it. The hierarchy has a natural termination at level n^* only if the gradient at that level reaches zero — no available $M^{(n^*+1)}$ can improve the learning mechanism further given current W. Whether such a fixed point exists, and whether it is reachable in finite time under realistic selection pressure, is open. Evolution [Hinton1987] and meta-learning [FinnEtAl2017, Thrun1998] are empirical candidates for $M^{(2)}$; whether anything instantiates $M^{(3)}$ in known systems remains unclear.

8. **Genetic encoding and the developmental sufficiency requirement.** The encoding hierarchy (Definition 8) describes a ratchet: proven behaviors promote toward L_0 as they meet the rigor threshold $\theta_0 \approx 1$ (Theorem 4). The question is what the asymptotic product of evolutionary gradient descent encodes [Schrodinger1944, MaynardSmith1995, Hinton1987, England2013].

   The framework does not resolve the content of genetic encoding — what specific mixture of M, (F_0, K_0), and developmental constraint DNA carries across all levels is an open empirical question. The claim is narrower: whatever genetic encoding carries must be *sufficient to develop the entity* — sufficient to bootstrap an organism capable of building (F, K) through interaction with W. The model is compatible with genetic encoding at any level of the hierarchy, in any proportion, provided the developmental output is a system that can run the gradient.

   What the framework does predict about the evolutionary rigor threshold: specific (F, K) are environment-dependent — a skill that reduces U in one environment may not transfer across the full distribution of environments the lineage will encounter. Content that clears θ_0 must generalize across that distribution. Developmental machinery — the programs, architectures, and gradient signals that build (F, K) — is more likely to generalize than specific content, because it is environment-agnostic. The framework predicts selection pressure favoring developmental generality over content specificity at the highest rigor levels; it does not predict that specific (F_0, K_0) is absent.

   This is consistent with the observation that we are not born with the context window of our parents: individual (K, F) — everything learned in one lifetime — clears θ_n but not θ_0. The inheritance mechanism transmits what is sufficient for development, not what was learned. The diversity of knowledge across individuals with similar genetic endowment is the expected outcome: the same developmental machinery operating on different environments produces different (K, F).

   Cultural transmission (language, writing, institutions) provides a parallel channel for (K, F) at lower fidelity but higher speed — $K_{collective}$ functioning as external inheritance (Corollary 15), serving a transmission function that genetic encoding cannot match for individual-lifetime content. The two channels operate at different rigor thresholds and different timescales.

   Formal questions that remain open: the precise allocation between M-encoding and (F_0, K_0)-encoding across evolutionary lineages; whether the modularity of genetic regulatory networks is derivable from the type structure of $M : (F, K) \times \mathcal{L} \to (F, K)$ at the evolutionary scale; and what determines θ_0 in terms of environmental distribution statistics.

9. **The AGI condition.** If $\mathcal{A}$ is necessary for genuine intelligence (Definition 7) and $\mathcal{A}$ cannot be imposed as an external training objective without becoming a simulation of the gradient rather than actual descent on it: what conditions are sufficient for $\mathcal{A}$ to emerge in an artificial system?

The framework does not resolve this. What it establishes is necessary: $\mathcal{A} > 0$ requires a real gradient — uncertainty that is genuine, not simulated — and $\eta_M > 0$ requires that the system's learning mechanism actually responds to it. Whether these conditions can be instantiated artificially, and what architectural or grounding requirements they impose, is an open question that the framework frames but does not answer.

10. **The next precipitation: normal form priority and extraction protocol.** The framework establishes that M_{res} is the unprecipitated cross product and that each precipitation reduces per-entity operational complexity. The open question is: given the current state of M_{res} and $K_{collective}$, which factor is most achievable to extract next, and what is the protocol for identifying and extracting it?

Priority criterion. "Most achievable" is under-defined by the framework. Natural candidates: (a) the factor whose precipitation threshold θ (Theorem 4) is closest to being met given current $K_{collective}$ — the factor requiring the smallest remaining evidential gap; (b) the factor producing the largest reduction in $|M_{\text{res}}|$ per unit of additional evidence; (c) the factor with the strongest cross-entity covariance signal in $K_{interaction}$ — where most entities' projections agree, suggesting genuine structure rather than projection artifact. These need not rank identically: the most evidentially ready factor may not be the one with the highest leverage. A principled priority function over these criteria does not exist in the framework as currently stated.

The identification signal. A partial answer via Theorem 11: convergence of independent agents on the same behavior pattern is evidence of an underlying precipitable factor — the pattern is the shadow of the factor in $K_{interaction}$. When many entities independently arrive at the same proposition from distinct starting points, that proposition likely grounds a precipitable dimension. The competing bounds problem (Open Question 6) may be directly connected: when multiple $f_i \in F$ compete at the same region of W, the competition may indicate an unnamed dimension whose precipitation would resolve the conflict. Competition at the action level is a signal of unresolved structure at the M level.

Extraction and verification. Necessary condition for a genuine new factor $F^{(k)}$:

$$I(F^{(k)}\,;\, w \mid F^{(1)}, \dots, F^{(m)}, K) > 0$$

— residual information gain after conditioning on all existing named factors. The implied protocol: (1) identify recurring propositions in $K_{interaction}$ across multiple entity subsets; (2) hypothesize the minimal factor $F^{(k)}$ whose naming would ground those propositions without derivation from existing named factors; (3) test at multiple entities — if independently adopting $F^{(k)}$ achieves $O(1)$ access to the propositions it grounds, the factor is real. If naming $F^{(k)}$ yields no access advantage — if the grounded propositions remain $O(|M_{\text{res}}|)$ — the candidate is a re-expression of existing named factors, not a genuine new dimension. The collapse case (Step 2b of Theorem 10) is a special instance: two apparent dimensions D_a and D_b are the same factor when naming one yields $O(1)$ access to what the other grounded.

This question is not merely theoretical: it is the operational question any entity with $\mathcal{A} > 0$ faces at the frontier of its M_{res}. The OKG framework described in Section 1 is an attempt to apply exactly this protocol to the domain of software engineering — identifying which actions have sufficient evidence to be precipitated as named factors and which remain in M_{res}, and which apparent distinctions between actions are the same underlying factor observed from different positions in W.

A limiting claim: single-class ceilings. There is reason to believe that a collective composed entirely of entities from a single sensing class — sharing the same σ-structure and therefore the same observable subspace of M — exhausts within-class precipitation at some finite point. Entities of the same class can jointly observe the dimensions of M accessible to their shared σ; once all cross-dimensional factors within that subspace have been extracted, further co-variance analysis within the class yields no new factors. The ceiling is the effective rank of the class's jointly observable subspace, which is strictly less than $\text{rank}(M)$ if dimensions exist that are accessible only to entities of a different class. Beyond this ceiling, further precipitation requires inter-class exchange — entities whose σ accesses dimensions outside the current subspace. This makes a testable prediction: the marginal precipitation value of adding a new entity to a homogeneous collective decreases and approaches zero; the marginal precipitation value of adding the first entity of a new class is non-zero regardless of collective size. The formal characterization of sensing classes, class-observable subspaces, and the inter-class exchange requirement is an open problem.